Middle Years

GEOGRAPHY

1 Understanding map basics

Dave Morrison

INSTANT LESSONS

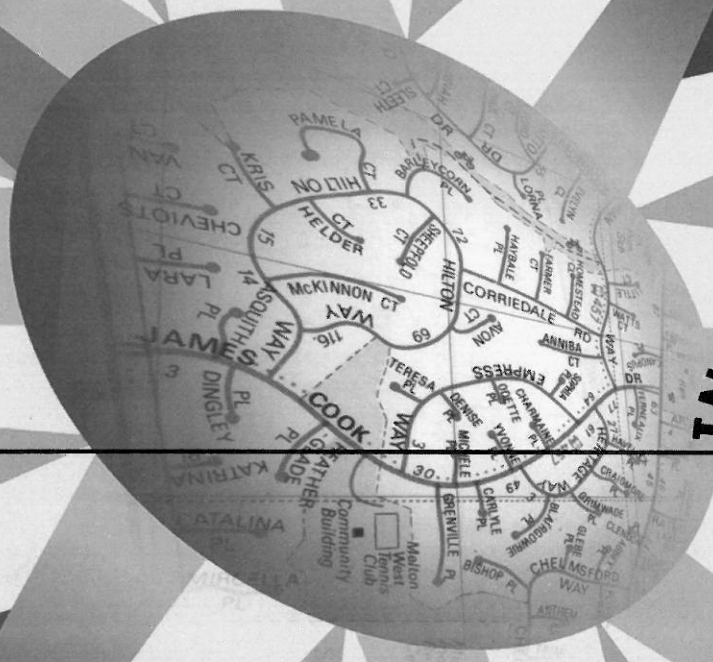
Quality ready-to-use resources

Middle Years

GEOGRAPHY

1 Understanding map basics

Dave Morrison



INSTANT LESSONS

© Learning Essentials Pty Ltd, 2004
Reprinted 2006, Blake Education

Copying of this book for educational purposes
The purchasing educational institution may only photocopy pages within this book in accordance with The Australian Copyright Act 1968 (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act.

It is mandatory, that ALL photocopies are recorded by the institution for CAL survey purposes.

For details of the CAL licence for educational institutions contact:
Copyright Agency Limited
Level 19, 157 Liverpool St
Sydney NSW 2000

The Act does not permit the purchasing educational institution to store any part of this book in a retrieval system, or transmit any part of it in any form by any means without the prior written approval of the publisher. All enquiries should be made to the publisher at the address below.

Publisher and editor: Susan Watson
Page design and formatting: Crackerjack Desktop Services
Cover design: Gail McManus Graphics
Maps and diagrams: Infographics Pty Ltd
Printed in Australia by Imprenta Pty Ltd
Reprinted in Australia by Ligare Book Printer 2006

Blake EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Rd
Clayton South VIC 3169
Ph: +61 3 9558 4433

ISBN 978 1 92561 035 2

Introduction

Instant Lessons in Middle Years Geography 1 is part of a two-book series that provides photocopiable worksheets to teach students essential geographic skills. They could be used with students from upper primary level through to year 9. In **Book 1**, students explore each topic on basic mapping skills through a series of information boxes, relevant maps and other graphics, definitions, and graded questions.

Most worksheets provide space for students to answer directly onto them, but several do require the use of a separate workbook. Topics 16–19 ask students to apply all the concepts. The two base maps in Topic 20 are used for activities in several different topics.

As the maps and symbols are not in colour, it would be a valuable additional activity to ask students to colour them if not already part of the worksheet. Using the appropriate colours on maps and symbols is another helpful learning technique.

Contents

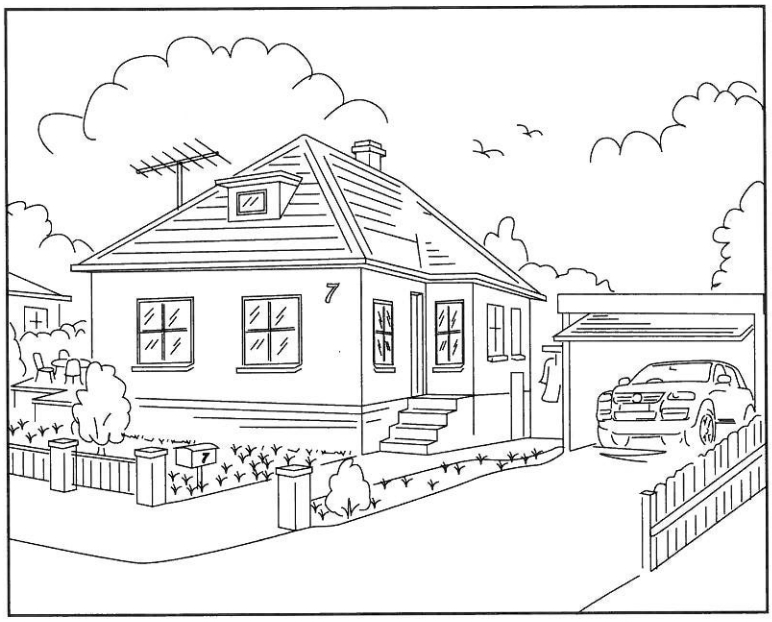
1 Different views 4
2 Mapping a plan view 6
3 Map symbols and key 8
4 Different types of map symbols 10
5 Which way to go? 12
6 Getting your bearings 14
7 The size of things on maps 16
8 Types of map scale 18
9 Maps at different scales 20
10 Measuring straight-line distance 22
11 Measuring non-linear distance 24
12 Measuring area on maps 26
13 Representing height on maps 28
14 Finding things on maps 30
15 From area references to grid references 32
16 Mapping a tribal group area 34
17 Mapping ancient Athens 36
18 Different views of a city 38
19 Multiple choice test 40
20 Base maps – Cape Beauty and Lake Mystery 42
Selected answers 44

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

1 Different views

There are different ways of viewing the same thing. Today we are going to look at three ways of viewing things. The three views are a **ground view**, an **oblique view** and a **plan view**. You can see a picture of Max Quickdrawer and Paula Mapmaker in a hot-air balloon. They will help us to learn more about these views.

Last week Max and Paula went on a hot-air balloon flight. The flight left from the park near Max's house. They floated over his house and landed about five kilometres away. They drew some different things they saw on the flight.

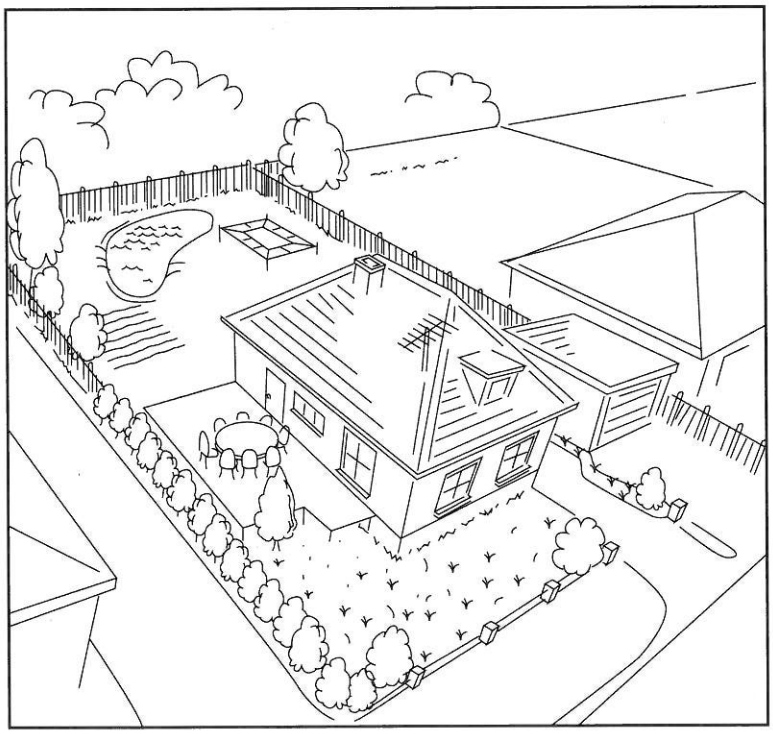


Ground view

This is Max's drawing of his house he did before the balloon flight began. When he drew it, he was standing on the street at the front of his house. It is a **ground view** of his house.

Some of things that you can see in this ground view are the front fence and the number on the house.

1 List 5 more things that are in the ground view.



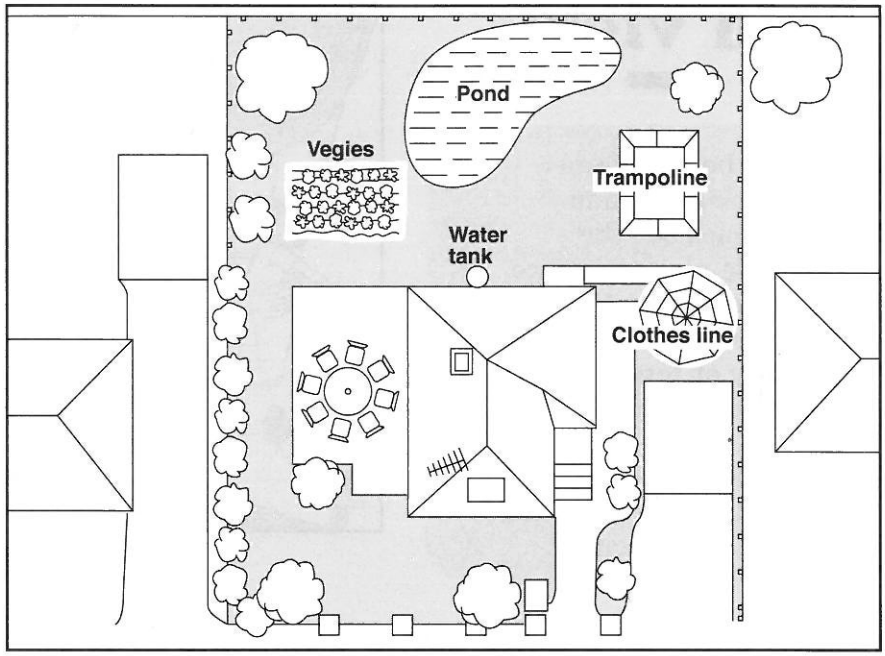
Oblique view

As the balloon started to rise Max quickly did another drawing looking at his house at an angle. Paula knows that this view is called an **oblique view**. This view is sometimes called an oblique aerial view as it gives a sideways view from up in the air.

2 List 5 things that you can see in this oblique aerial view.

3 List 3 things that are not in the ground view.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1



DEFINITIONS

ground view shows what something looks like from the ground. It is a side-on view.

oblique view shows what something looks like from an angle. It is a view that is partly above and partly to the side.

plan view shows what something looks like from directly above. It is a bird's-eye view.

Plan view

When the balloon was floating slowly over his house Max did this drawing of the house from a different angle. This view looks down from directly over the house. It is called a **plan view**.

4 List 5 things that you can see in this plan view.

_____

_____

_____

5 List 3 things that are not in the ground view.

_____

_____

6 List 3 things that are not in the oblique aerial view.

_____

_____

7 Why is a plan view often called a bird's-eye view?

_____

_____

People use these three different views – ground view, oblique view and plan view – to look at many different things in our surroundings. People make drawings and maps from these views.

8 Think back to when you were younger. Think of a bucket, spade and ball that you took with you when you played on the beach. When you arrived at the beach you put the bucket and ball on the sand and stuck the spade straight into the sand.

- **a** In your workbook, draw a ground view of the bucket, spade and ball showing how they look when you lie on the sand to do your drawing.
- **b** Now do a plan view showing how the bucket, spade and ball look if you stand over the top of them.
- **c** If you were on your knees you would have an oblique view. Draw an oblique view of the bucket, spade and ball.
- **d** Label your drawings with the correct view for each drawing.

9 Put a tick next to the view that is best to see these things in Max's drawings:

Front door	☐ ground	☐ oblique	☐ plan
Trampoline	☐ ground	☐ oblique	☐ plan
Vegie patch	☐ ground	☐ oblique	☐ plan
House number	☐ ground	☐ oblique	☐ plan
Size of house	☐ ground	☐ oblique	☐ plan
Backyard	☐ ground	☐ oblique	☐ plan
Front fence	☐ ground	☐ oblique	☐ plan

10 In your workbook, draw these things in ground view and then in plan view.

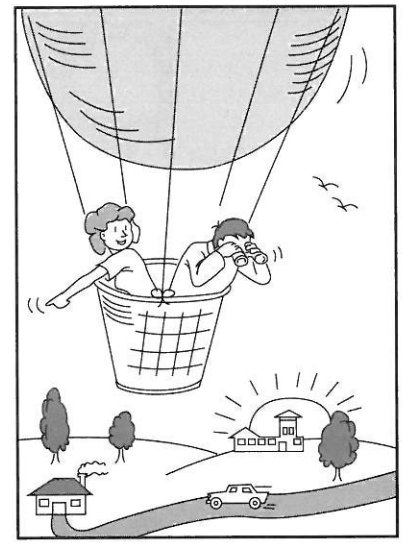
- **a** a bicycle
- **b** a tin of paint
- **c** a ruler
- **d** a book

11 Choose three more things yourself and draw them in your workbook, in ground view and then in plan view.

© Blake Education 2006

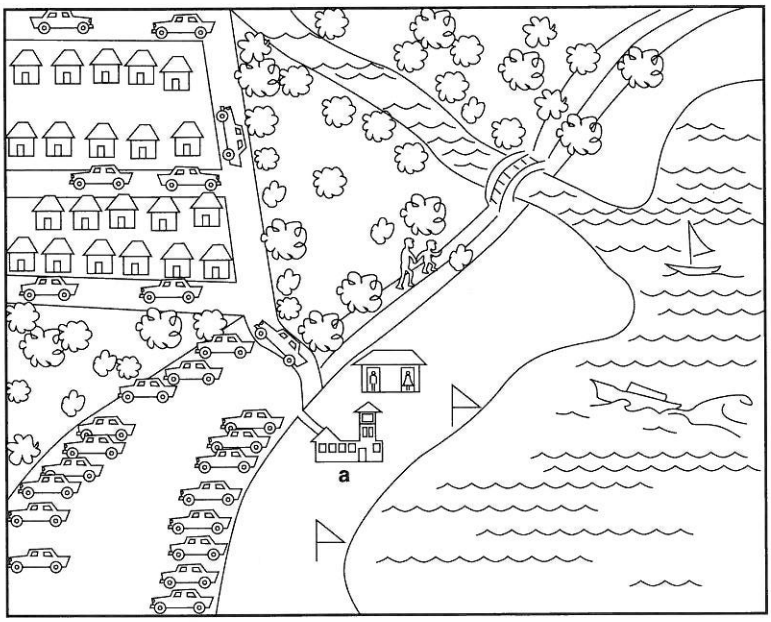
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

2 Mapping a plan view



Maps are drawn as if you were looking at the area from above. They show a plan view. A person drawing a map cannot show everything on the map. They pick out the important things to show. A map is a simple plan of a place from directly above. A map could be of a small area like a classroom or it could be of a large area like the world. A map normally has a **title**, a north pointer, a **key** or **legend** of the symbols used and a **scale**. Most maps are enclosed by a square or rectangular border.

Last week Max Quickdrawer and Paula Mapmaker went on a hot-air balloon flight. The flight left from the park near Max's house. They landed at a beach about five kilometres away.

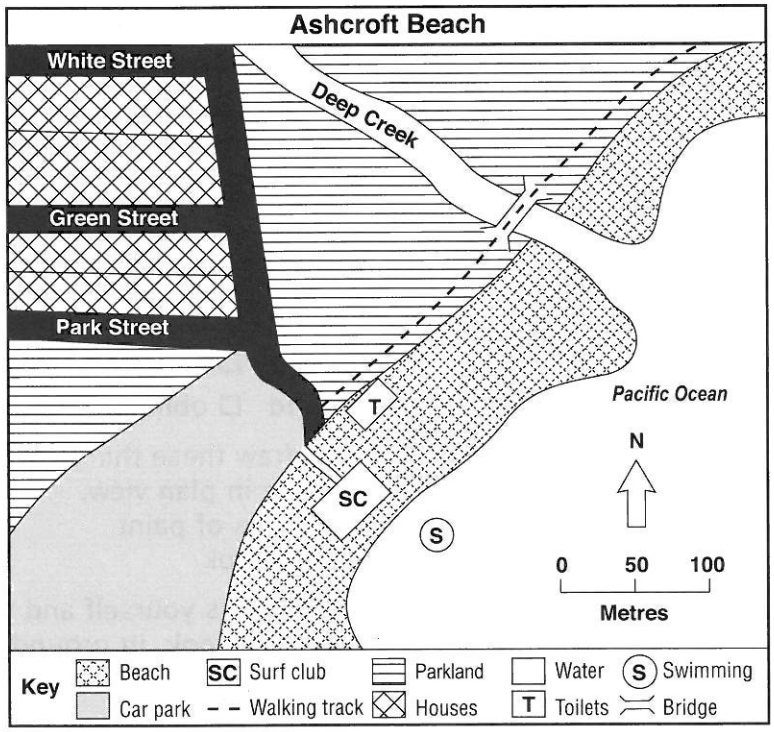


Plan view drawing of Ashcroft Beach

Max is a good drawer and this is his drawing of Ashcroft Beach.

1 There are many different things in Max's drawing. See if you can find these things. Write the letter on his drawing where you find them. The first one has been done for you:

a surf club
b road
c house
d another house
e beach
f boat
g bridge
h car on road
i 17 cars in the car park



Map of Ashcroft Beach

Paula is good at making maps. This is her map of the beach.

Maps show a large area like the beach in a small space. This map is just 10 centimetres wide. The real-life distance across the map is 500 metres.

A map shows the main features of an area, not every detail.

2 Which view does this map show?

☐ ground
☐ oblique
☐ plan

Explain _____

_____

_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

2 Mapping a plan view (continued)

If you look at Max's plan view drawing and Paula's map you can see that they both cover the same part of the beach.

Some of the things that they both show are the surf club, the beach and the roads.

Paula has given her map a title. The title tells us what the map is about. The title of this map is Ashcroft Beach.

3 Draw a ring around the title of the map.

4 Write the title of this map here.

_____

5 Why do you think the words 'Map of' do not appear in the title?

_____

Elements of maps

A map uses colours, shadings and **symbols** to tell us about many of the things on it. Paula has used a solid black colour for the roads. She has used a light grey for the car park.

Paula uses a **key** so we know what the colours, shadings and symbols mean. A key is an important part of a map. It helps us understand what is shown. On Paula's map the key is near the bottom of the map.

6 On the map of Ashcroft Beach what does the shading XXXXX mean?

_____

7 Draw the symbol for the safe swimming area.

DEFINITIONS

title names the place a map is representing

scale is the relationship between a unit of length on a map and the corresponding length over the ground

direction is the location based on the points of a compass

symbols are the markings used to represent features or objects

key explains the symbols used in a map. It is also called a **legend**.

North, south, east and west are **directions**. The sun rises from the direction called east and the sun sets in the direction called west. Paula has drawn an arrow with an N to show which way is north.

She has also drawn a scale so we can work out the distances shown on her map.

8 On the map, use black to colour in the arrow that shows which direction is north.

9 Fill in the missing word using one of these three words. (side, top, bottom).

The north arrow points to the _____ of the page.

You will notice that the map does not show things in the same way as they are in Max's drawing.

One example is the houses. In the drawing, Max drew them to show their real shape. The map uses the same symbol for the group of houses, as Paula just wanted to show that there were many houses along the road.

Now answer questions 10–14.

10 Find 5 other things that look quite different on the map from the drawing.

_____

_____

11 Find some things that are in the drawing but not on the map.

_____

_____

12 Find some things that are on the map but not in the drawing.

_____

_____

13 Most maps use colour. Choose suitable colours for the sea, the car park and the sandy beach. Colour in the map with these colours. Make sure you colour in the key as well.

14 We found out earlier that you can draw a map of a small area like classroom or a big area like the whole world. Make a list of other maps you have seen.

_____ _____

_____ _____

_____ _____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

3 Map symbols and key

A map uses **symbols** to represent features or objects on the ground. A map is not a drawing of how things really look. Most maps use many symbols. So we can easily understand the map, all the symbols used are grouped together in one place. This is called the **key**. Another word used instead of key is legend. A key on one map could be called a legend on another map.

Max Quickdrawer and Paula Mapmaker help us learn about maps. Last week Max and Paula went on a hot-air balloon flight. The flight left from the park near their school. They floated over their school and landed about five kilometres away at Cape Beauty.

NOTE TO TEACHER

This worksheet uses the map of Cape Beauty on page 42. To complete this worksheet, students will need a copy of this map.

Paula drew the map of Cape Beauty (on a separate sheet) as they floated by. Paula used symbols on her map to show clearly some of the features that she saw. The symbols help us to get a good idea of what Cape Beauty looks like.

A symbol is not a drawing of a feature. It is a representation. It needs to be simple. It is good if we know what the symbol represents just by looking at it.

Paula has drawn a key of all the symbols used on her map. She wanted to be sure that we knew the meaning of each of the symbols.

1 Which building is next to the hospital?

_____

2 What feature is at the end of the cape?

_____

3 Beach Parade is one of the main roads. Make up names for the other main roads and write them on the map.

4 How many shopping centres are shown?

_____

5 Which symbols are easy to recognise?

_____

_____

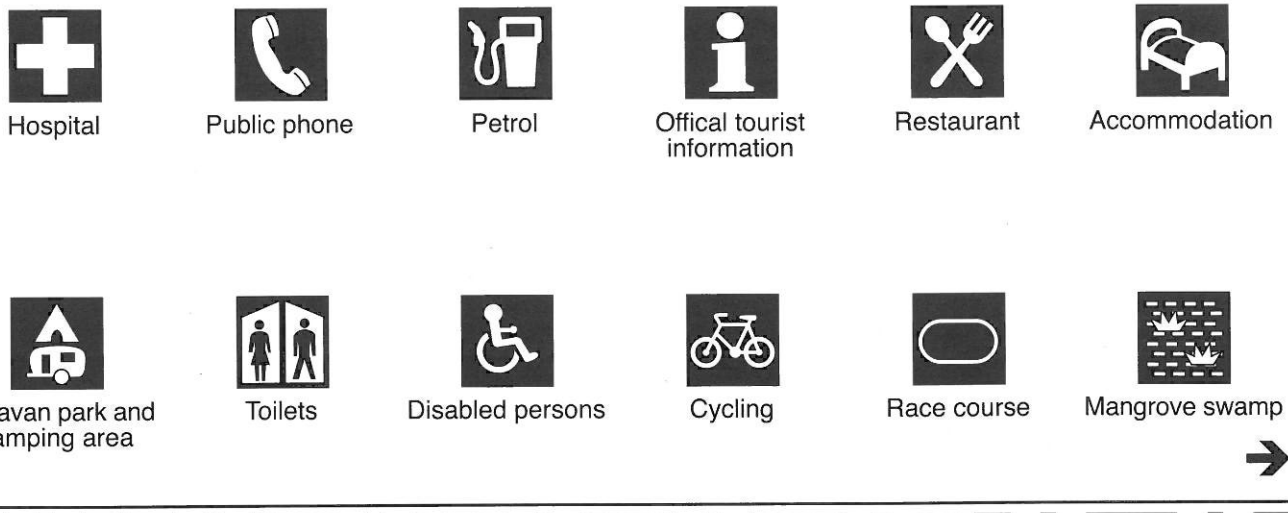
6 Which symbols are hard to recognise?

_____

_____

7 There are no picture symbols for the housing areas. These are all the areas along the roads that are not used for anything else. These are the residential areas. Why aren't picture symbols used?

Paula has a collection of symbols that she often uses when she is mapmaking. As well as the ones on the Cape Beauty map, these are part of her collection.



© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

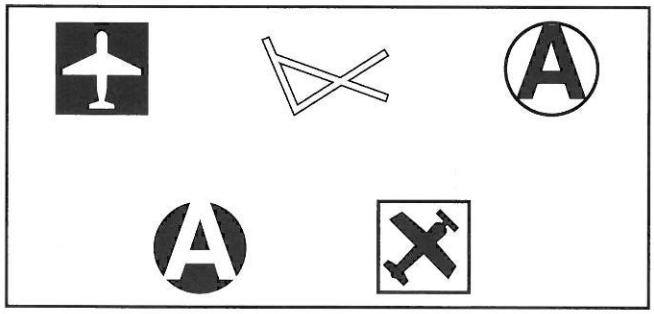
8 In your workbook or on a sheet of paper copy each of Paula's symbols from the previous page. Underneath each symbol write a sentence about it. The first sentence has been done for you.

My mum went to hospital when she broke her leg.

Different symbols for the same thing

There are many ways to draw a symbol for most features.

Paula collects maps that other people have drawn. She has noticed there is more than one symbol used for airports. Here are some of them.



Coloured symbols

Many maps are coloured. Sometimes the symbols on maps are different colours for different features. For example, blue is often used for water and green for parks.

9 Choose 6 features from Paula's list of symbols. Make your own new symbol for each of them. Add colour. Remember that a symbol is not a drawing of a feature. It needs to be simple and it is good if other people know what it means just by looking at it.

Other colours on maps

Using colours over an area is also a way of representing different features. Cape Beauty has these areas that could be represented by colours:

- residential houses along the streets
- reserve (natural parkland)
- ovals
- sea
- sand along the beach.

10 Choose suitable colours for each of the areas in the list above. Colour these areas on the map with your chosen colours. Colour the key for these features and add a new one for Sea.

11 Where would you like to live in Cape Beauty? Why?

12 Where wouldn't you like to live in Cape Beauty? Why?

13 In your workbook map your classroom using symbols for single objects such as the teacher's table, board, bookshelf and pot plants. Use different colours to show large areas such as student desks, windows and the outside corridor. Add a title and a key to your map. (Hint: your map does not have to be to the correct scale.)

LETTERS AS SYMBOLS

Letters can be used as symbols. Letters can also be coloured.

DEFINITIONS

symbol used on maps to represent features on the ground

key contains the collection of symbols used on a map. It is also called a **legend**

residential is the area mainly made up of places where people live, such as houses and flats

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

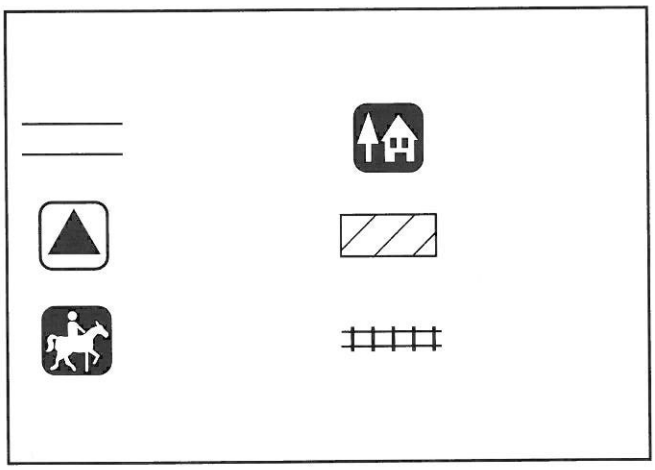
4 Different types of map symbols

Symbols are used on maps to identify features. Symbols allow a map to show lots of information in a small place. Symbols are drawn as lines to show features like roads, rivers and boundaries. Symbols can be drawn as points to show buildings, mountain peaks and capital cities. Symbols can be drawn as areas to show parks, forests and land-use zones. On most maps, colour is used to identify different groups of symbols.

The key or legend shows all the symbols used on a map and their meanings. The key normally shows symbols in groups. They can be **linear**, **point** and **area symbols**. The features can be natural like rivers, mountain peaks and wetlands or built areas like railways, schools and urban areas.

NOTE TO TEACHER

This worksheet uses the map of Lake Mystery on page 43. To complete this worksheet, students will need a copy of this map.



1 Refer to the map of Lake Mystery provided with this worksheet.

a Use the key to label the symbols shown above. Circle them on the map.

b Draw the symbols used to identify each of these features:

Lookout		Walking track	
Parking		Airport	
Picnic area		Lake	

c Jacinta left the lookout near the village and walked to the medical clinic. List the features that she passed on her walk.

_____

_____

_____

2 There were some features that were left off this map. Create symbols for these and add them to the map and the key.

a There are new toilet blocks at both lookouts.

b There is a new school that has been built just north of the police station.

c A restaurant will be built at the western corner of Lake Highway and Lake Mystery Drive.

Linear symbols

Linear symbols are those shown by lines on a map. The lines may have some width. The width may also have meaning. In a street directory different types of roads or highways will have different widths or colours but they will still all be linear symbols. The lines could be solid or broken or have various colours to represent different meanings.

3 List 3 linear symbols in the map of Lake Mystery.

_____

_____

4 List 5 more linear features that may be shown on a map. Use your atlas if you need to.

_____

_____

_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Point symbols

A **point symbol** locates the particular position of a feature. The symbol could just be a dot or more meaningful such as an aeroplane symbol for an airport. The size of the symbol generally does not represent the size of the feature.

5 List 3 point symbols in the map of Lake Mystery.

_____

_____

6 Create a point symbol for:

a a skateboard park ☐

b a rock music venue ☐

c a cinema ☐

Area symbols

An **area symbol** shows the size or shape of an area. Examples are the sea, forests and parkland. Colour or shading is often important in area symbols.

7 List 3 area symbols in the map of Lake Mystery.

_____

_____

8 Choose colours for the area symbols you listed in question 7 and apply them to the map.

9 Draw and colour an area symbol for:

a wetlands ☐

b desert ☐

Which symbol to use?

The type of symbol used often depends on the size of the area shown in the map.

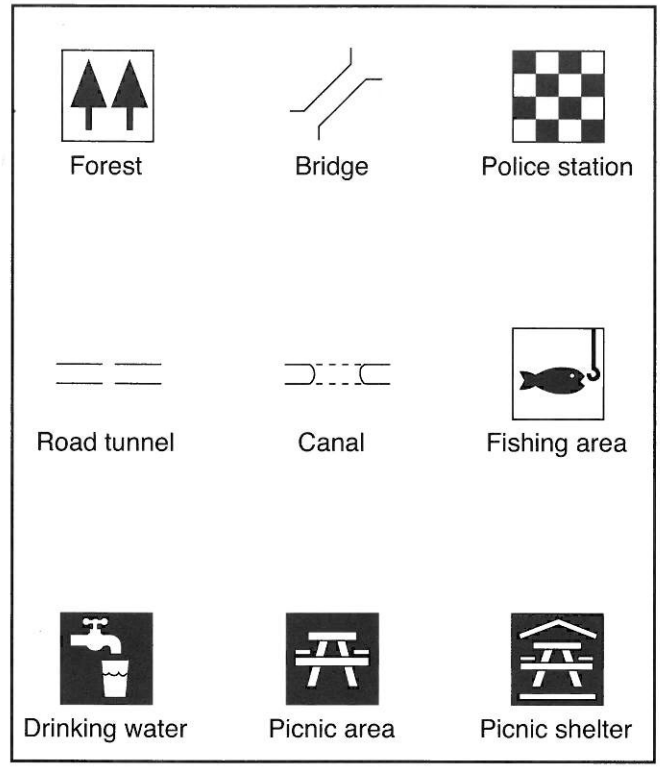
In a map of a whole town, a school might be shown using a point symbol such as a dot. In a map showing a smaller local area, the shape of the school ground could be shown. It could be coloured bright yellow if that was used as the area symbol colour for schools.

DEFINITIONS

linear symbol forms a line such as a street or a creek

point symbol shows a specific location such as a police station, a hospital or a bridge

area symbol covers an area like the sea, parkland or forest



10 Complete this table by listing each symbol above in the correct column.

Linear	Point	Area

11 Which of the features identified by area symbols could be point features on a map showing a much larger area?

_____

_____

_____

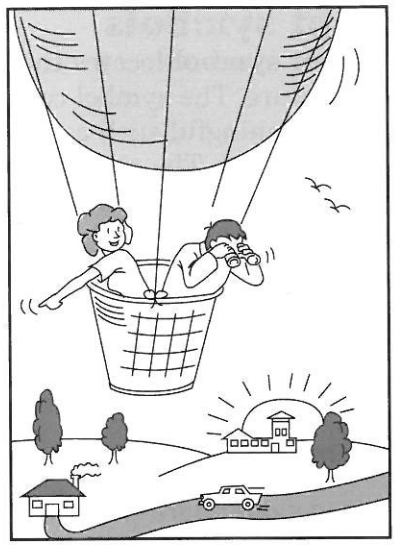
© Blake Education 2006

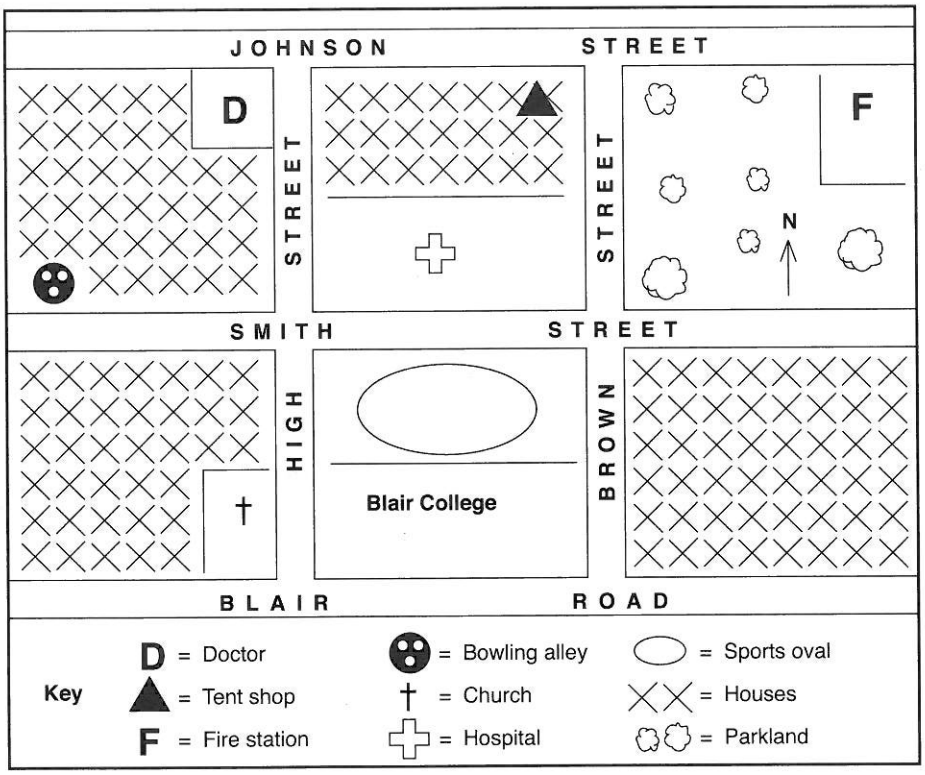
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

5 Which way to go?

The four main **directions** are North, East, South and West. One of the best ways of working out directions is to use a **compass**. The needle on a compass always points to the north. You can use directions to find your way from one place to another. On a map you can use directions to find out where one point is compared to another.

Max Quickdrawer and Paula Mapmaker will help us learn about maps. Last week Max and Paula went on a hot-air balloon flight. They took their compass with them. They floated over their school and passed over Blair Town.





This is the map Paula drew of part of Blair Town. It shows the school called Blair College and some of the surrounding streets and buildings.

1 List 5 features other than the college and streets that you can see on the map.

_____

_____

_____

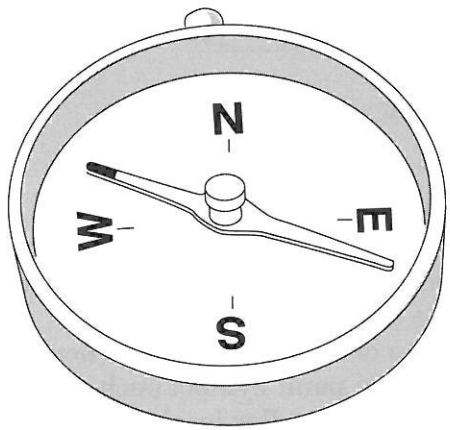
_____

_____

A compass

Max and Paula decided to work out the **direction** of their flight. Their **compass** shows the four main directions: North, South, East and West. These directions are also known as the **cardinal** points of a compass. They are shown on the compass to the right by their first letter.

The compass has two main parts. There is a magnetic needle that can move freely. It always points to the north. And there is the base that has the first letter of each of the four main directions. Paula shows Max how to find north by lining up the N on the base with the north pointing magnetic needle. Then the E is pointing east, the S is pointing south and the W is pointing west. These main four directions are called the cardinal directions.



2 On the compass above write in the full direction name next to the letters.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Many people remember the cardinal directions using a special phrase called a mnemonic. This first letter of each word is used to recall the facts to be remembered. One of these phrases is 'Never Ever Swallow Whales' which helps you remember North, East, South, and West.

3 In your workbook make a drawing showing why you should **N**ever **E**ver **S**wallow **W**hales.

4 Make up your own mnemonic using 4 words that begin with:

N _____

E _____

S _____

W _____

Where is North?

Maps usually have a north pointer to show the North direction. It can just be an arrow. It may have an N on the point of the arrow. It may be a complete compass that shows other directions as well.

Maps are usually drawn so north is at the top of the map but not always. The north pointer shows the orientation of the map.

5 If North is at the top of the map, which direction is at:

a the left side of the map?

_____

b the bottom of the map?

_____

6 Use the map on the previous page to answer these questions:

a Max noticed that High Street goes north from Blair College. If you walk north up High Street and turn right at Smith Street and walk past the sports ground, you would then be going towards the east or in an easterly direction. At Brown Street, if you turn around and go back down Smith Street towards High Street in which direction would you be going?

_____

b If you were at the corner of Smith Street and High Street, what is the direction to Blair College? _____

c What direction is the doctors from the tent shop? _____

d What direction is the fire station from the tent shop? _____

e Name at least 2 places that are north of Blair College. _____

DEFINITIONS

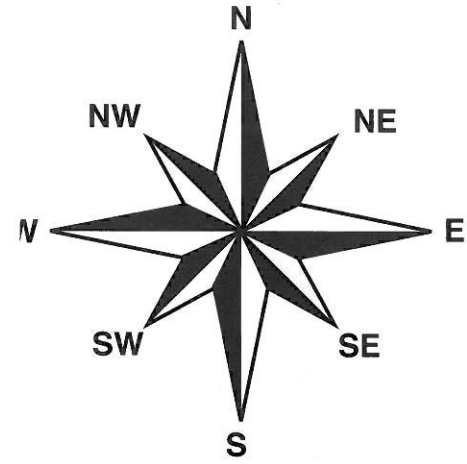
direction is the way from one point to another

compass is an instrument for finding directions. It uses a magnetised bar or needle turning freely on a pivot and pointing in a north-south direction

cardinal means main or of basic importance.

f If you were at the bowling alley what is the direction to the doctors? Hint: If you do not know the answer, come back to this question later. _____

The eight-point compass



As well as showing the cardinal points, an 8-point compass rose shows directions halfway between them. North-east is half way between North and East and is shown as NE on the compass. South-east is half way between South and East and is shown as SE on the compass.

7 What is the direction shown by NW on the compass? _____

8 Use Paula's map again:

a What direction is the bowling alley from the doctors? _____

b Name 2 things that are south-east of the bowling alley. _____

c Walk west along Blair Road to Brown Street. Walk north to the next street. Walk north-east until you find a building. What is it? _____

d Choose a secret location on the map and write the directions to locate it.

e Swap questions with a partner and find your partner's secret location.

9 Look at the drawing of Paula's compass. Is the compass needle pointing to the north? Explain your answer.

_____

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

6 Getting your bearings

A **bearing** is the angle measured in degrees between North and the direction of an object. There are 360 degrees in a circle. The four cardinal points are 90 degrees apart.

The symbol ° is used as an abbreviation for the word degrees. A bearing of 90° is **due** East, 180° is due South and 270° is due West.

Bearings are used in **navigation** particularly in air and sea navigation.

DEFINITIONS

bearing is the direction defined by the angle measured from north

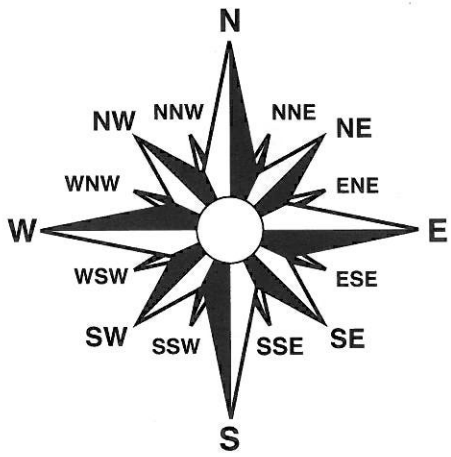
due means directly or exactly

compass rose is a full circular compass

navigation is finding the way from one place to another

The 16-point compass

The **compass rose** below is a 16-point compass. It shows the letters that represent the 16 different directions.



1 If NE means North-east what is the meaning of:

a SW _____

b SE _____

c NW _____

2 NNW on the 16-point compass are the letters for the direction of north-north-west. List all 16 directions.

3 If the point north of north-east is shortened to NNE what are the short forms of:

a south of south-east _____

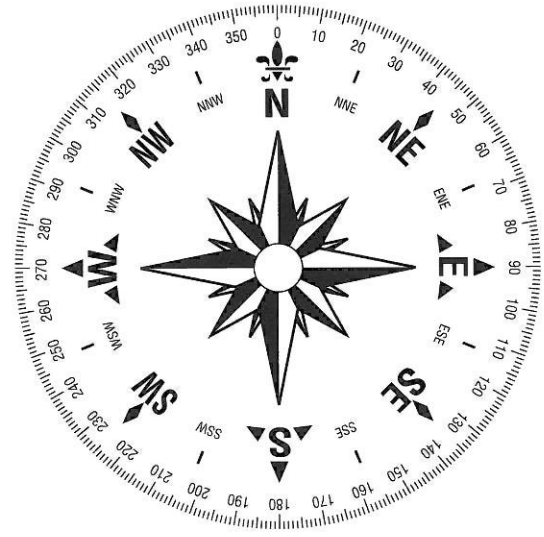
b west of north-west _____

c north of north-west _____

4 What direction is halfway between east of southeast and south of southeast? _____

Bearings

Bearings have been added to the compass rose. Bearings are measured in degrees clockwise starting at the north point. There are 360 degrees (or 360°) in a circle. The east point is at a bearing of 90°.



5 What bearing is:

a west _____ **d** ESE _____

b south-east _____ **e** WSW _____

c north-west _____ **f** NNW? _____

6 Why do you think bearings are more important in air and sea travel than travelling by road or train?

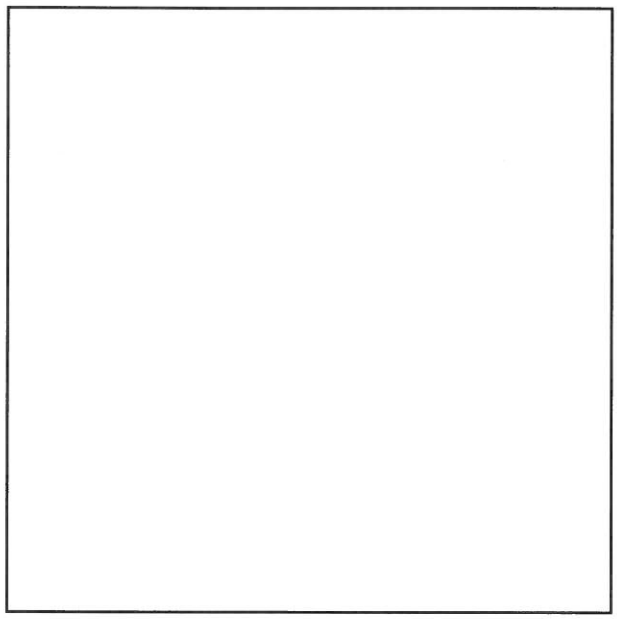
_____

_____

_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

7 Design your own 8-point compass and show the bearings of each direction point.

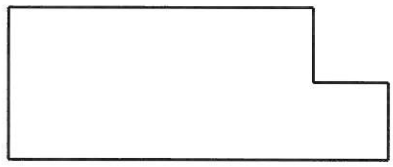


8 The table below shows the bearing taken and the distance travelled by some scouts. Each row shows the change in bearing.

a From the Start point, add an arrow on each of the parts of the track. The table tells you that first direction is east.

b Complete the table for this track taken as shown in the diagram.

Start



Bearing	Distance	Length of line on map	Direction
90°	4 km	4 cm	East
180°	1 km	1 cm	
90°			

9 Draw the tracks taken by these scouts. (Hint: Refer to the compass rose showing bearings. You can fill in the direction column to help.)

a

Bearing	Distance	Length of line on map	Direction
180°	3 km	3 cm	
270°	4 km	4 cm	
45°	5 km	5 cm	

b

Bearing	Distance	Length of line on map	Direction
90°	3 km	3 cm	
180°	3 km	3 cm	
270°	1 km	1 cm	
0°	2 km	2 cm	
270°	2 km	2 cm	
0°	1 km	1 cm	

c Did both groups of scouts get back to their starting point? _____

10 Sam is standing in the middle of his farm. He looked around and these are some of the things he can see.

At a bearing of:

- 0° is a windmill
- 90° is a vineyard
- 20° is a tractor
- 340° is his house
- 270° is a forest
- 180° is the stable for his horses.

Create a drawing or a map in your workbook to show Sam and his farm. Choose your own distances between Sam and what he sees.

11 Give instructions for getting from your desk, out the classroom door, and to the main school gate. Use bearings.

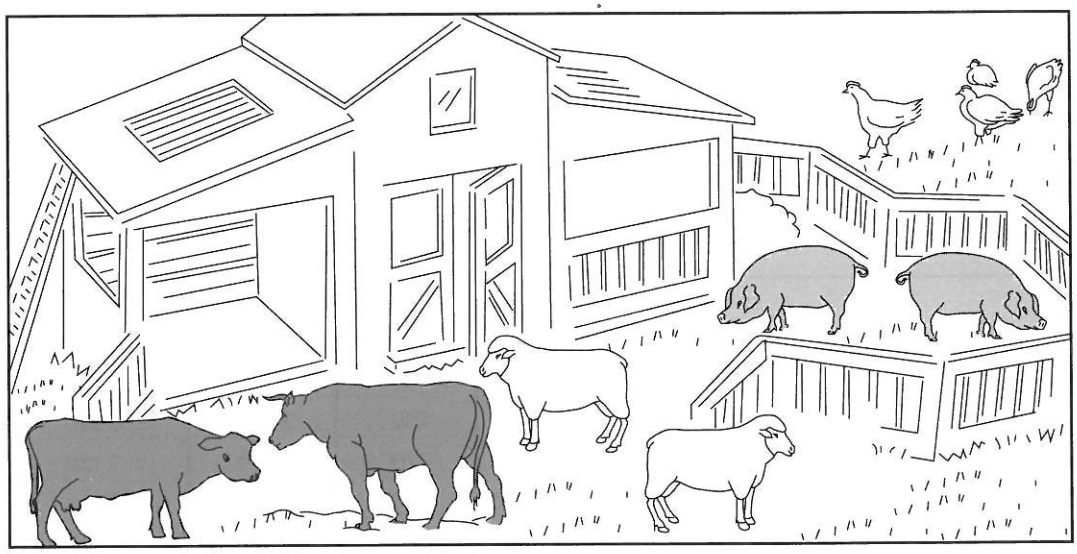
© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

7 The size of things on maps

Maps show features of the land smaller than they really are. To draw a map the area must be reduced or scaled down to fit on the page. If not, the map of your town would have to be the same size as your town. The **scale** of a map tells you what distance on the map represents what true distance on the ground.

Max Quickdrawer and Paula Mapmaker help us learn about maps. Last week Max and Paula went on a hot-air balloon flight. They tried to think of a way to put some of the things they saw on a map. They thought of their model railway and farm.



Scaling things down

The animals in Paula's farm above look just like real animals except they are a lot smaller than the real thing. Maps use the same idea. You cannot draw a map to be the real size. You have to scale it down just like Paula's animals.

To work out the scale we need to know how much smaller is the size of Paula's sheep than a real sheep. Paula's sheep is 2 centimetres long. If a real sheep is 100 centimetres long then the scale would be 2 to 100, which is a scale of 1 to 50. Each one centimetre of the model sheep represents 50 centimetres in reality.

To find out the real size of the other animals we need to measure the model and multiply the answer by 50.

We can state this as a **ratio scale** of 1:50.

We could create a **linear scale** like this:

1 cm (on map) = 50 cm (on ground)

0 50 100 150 200

1 Find out the length of another animal that is 3 centimetres long using the same scale of 1:50.

Calculation

1 cm = 50 cm

3 cm = 3 x 50 = _____ cm

There are 100 cm in 1 m

The animal is _____ m long.

2 What could this animal be? _____

3 Look at the drawing of Paula's model farm. Remember the animals are drawn at a scale of 1:50.

a What is the length of the sheep in the drawing? _____

b What is the real life length of the real sheep? _____

c What is the length of the cow in the drawing? _____

d What would be the real length of the cow? _____

e What is the height of the cow in the drawing? _____

f What would be the real height of the cow? _____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

DEFINITIONS

scale is an indication of the relationship between distances on a map and the actual distances

ratio scale is scale expressed as a proportion

linear scale is scale shown on a line of measurement

distance is the space between two objects or points

4 Use the same scale as Paula's farm to draw a horse on a separate sheet of paper.

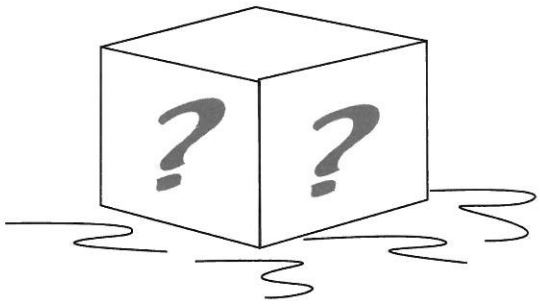
a You can draw a horse that in reality is 200 cm in height and 250 cm long or make up your own size horse.

b Put the scale used on your drawing.

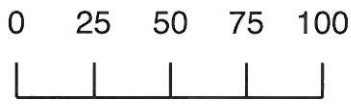
c Complete this table:

Horse	Height	Length
In drawing		
In real life		

5 This is a Mystery Box drawn at the scale of 1 to 50. The Mystery Box is really 100 centimetres high and 100 centimetres wide.



a Draw the Mystery Box using this scale.



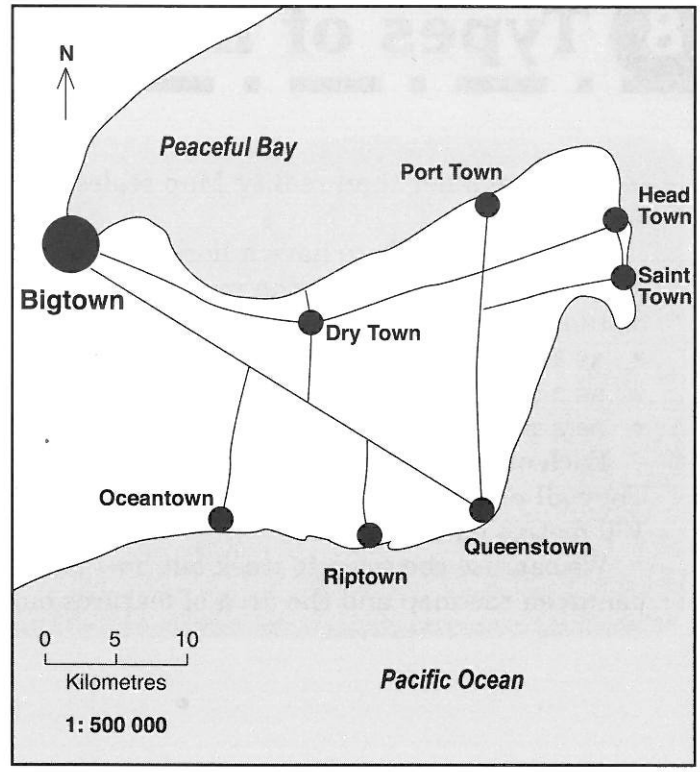
b Draw the Mystery Box at a scale where 1 centimetre on the drawing represents 1 metre in real life.

c Draw the Mystery Box using a scale of 1:10.

d Which of these scales do you think is the best? _____

_____

e Use your imagination. What do you think is inside the Mystery Box?



A map needs to show a scale. It shows what **distance** on the map represents the distance on the ground. On this map the scale is 1:500 000.

One centimetre on the map represents 500 000 centimetres on the ground. This is 5000 metres or 5 kilometres.

We can use the scale to work out how far it is between points on the map above. The distance on the map between Oceantown and Riptown is 2 cm. As 1 cm on the map represents 5 km, we can work out that Oceantown and Riptown are really 2 times 5 or 10 km apart.

6 Dry Town and Port Town are 3 cm apart on the map. How far apart are they on the ground? _____

7 Bigtown and Queenstown are 7 cm apart on the map. The distance between them is

_____

8 Trace the map and add another section to the west at the same scale. You can decide what to show in your map or you can include features such as:

- Lake Watson
- Big River
- and some more towns and roads.

9 In your workbook, draw a map to scale of one of the following:

- a desk
- a table
- a house block showing the buildings such as house, garage and shed.

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

8 Types of map scale

Maps are smaller than reality. Map scales describe how much smaller.

Every map needs to have a horizontal scale. The scale shows what **distance** on the map represents the real distance on the ground. Scale can be shown in three ways:

- as a **sentence**
- as a **linear scale**
- as a **ratio scale**.

Each of these scales shown below means the same thing. They all mean that 1 centimetre on the map represents 100 metres on the ground.

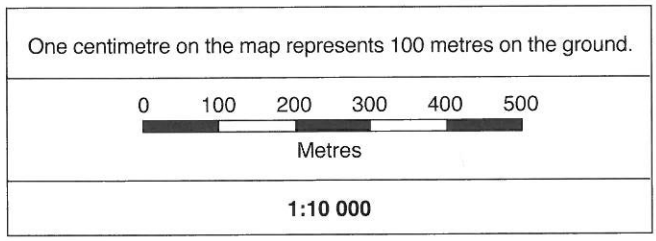
We can use the scale to work out how far it is between points on the map and the area of features on it.

DEFINITIONS

distance is the real space between two features or places on a map

ratio scale is indicated by a ratio of one unit on the map to some number of the same units on the ground, for example 1/10 000 or 1:10 000

RF stands for representative fraction scale, which is another name for a ratio scale



Sentence scale

A sentence scale states the scale of the map in the form of a statement. For example, one centimetre on the map represents 100 metres on the ground.

If 10 kilometres on the ground is shown as 1 centimetre on the map the sentence scale would be:

One centimetre on the map represents 10 kilometres on the ground.

1 Write each of the following as a full sentence scale:

a 1 centimetre represents 1 kilometre

b 1 cm = 100 km

c 1 kilometre is shown as 1 centimetre

2 A map has a scale where 1 centimetre on the map represents 1 kilometre on the ground. What is real distance between points this far apart on the map?

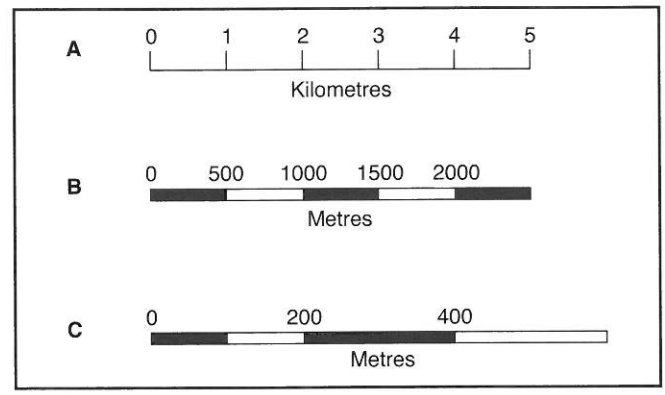
a 2 centimetres _____

b 5 centimetres _____

c 10 centimetres _____

Linear scale

There are different styles of linear scales.



A linear scale is also known as a bar scale. The linear scale **A** above shows 1 centimetre on the map represents 1 kilometre on the ground.

3 Convert **B** and **C** above to sentence scale.

4 Draw a linear scale showing that 1 cm on the map represents 1000 km.

5 Use the linear scale you have drawn in question 4. How far apart on a map would be places that are this far apart on the ground?

a 2000 kilometres _____

b 100 kilometres _____

c 4000 kilometres _____

d 1 kilometre _____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Ratio scale

The ratio scale is expressed as a ratio in the form such as 1:10 000 . This means that the size of objects on the map is $\frac{1}{10\,000}$ of their size on the ground. Another way of stating this is 1 unit on the map represents 10 000 units on the ground.

The unit used is not important. If the ratio scale is 1:10 000, 1 millimetre represents 10 000 millimetres and 1 centimetre represents 10 000 centimetres.

The ratio is also called the 'representative fraction' or 'RF' of the map.

6 Write the ratio scale of a map that shows distances as $\frac{1}{500}$ of the real distance. _____

7 Write the ratio scale of a map that shows distances as $\frac{1}{200\,000}$ of the real distance. _____

8 On a map with a ratio scale of 1:100 000 what is the real distance in centimetres between 2 points that are 1 centimetre apart on the map? _____

9 On a map with a ratio scale of 1:100 000 what is the real distance in glonks between 2 points that are 1 glonk apart on the map? _____

People often use a map to calculate a distance on the ground. If you understand ratio scales, it is easy to do the calculation. See the box at the top of the next column to learn how.

10 For a map with a scale of 1:10 000 what are the real life distances between two points that are this far apart on the map:

a 3 centimetres _____

b 8.5 centimetres _____

c 0.5 of a centimetre _____

11 Calculate the missing figures in this table.

Ratio Scale	Distance on the map	Distance on the ground
1:100	4 centimetres	
1:250 000	5 centimetres	
	8 centimetres	80 000 centimetres
1 10 000		30 000 centimetres
1:10 000		1.5 kilometres

CALCULATING DISTANCES USING RATIO SCALE

We know the distance between 2 points on a map, with a scale of 1:100 000, is 7.5 centimetres. What is the distance on the ground?

Calculation

1 centimetre represents 100 000 centimetres

so 7.5 centimetres represent 7.5 times 100 000 centimetres

= 750 000 centimetres.

Convert to metres by dividing by 100 as there are 100 centimetres in a metre

= $\frac{750\,000}{100}$ metres = 7500 metres.

Convert metres to kilometres by dividing by 1000 as there 1000 metres in a kilometre

$\frac{7500}{1000}$ kilometres = 7.5 kilometres

12 Look at the linear scale examples earlier in this worksheet. Express the scale for **A**:

a as a representative fraction _____

b in sentence form _____

_____

c Which type of scale is the best? Give reasons for your choice.

_____

_____

_____

_____

_____

_____

_____

_____

13 a Define 'horizontal scale'.

_____

_____

b What are the three types of horizontal scale on maps?

_____

_____

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

9 Maps at different scales

The terms 'large-scale maps' and 'small-scale maps' are often confused. 'Large scale' refers to maps on which objects are relatively large and 'small scale' to maps on which objects are relatively small. So small-scale maps cover a much larger area than large scale maps.

When Max Quickdrawer and Paula Mapmaker flew in a hot air balloon from a park near their home to the beach five kilometres away, they flew at about 300 metres above ground level. They saw things at a large scale. This week Max and Paula took a plane flight from their home city to Palm Island off the coast. The plane flew at a height of 3 kilometres. They saw things at a small scale.

DEFINITIONS

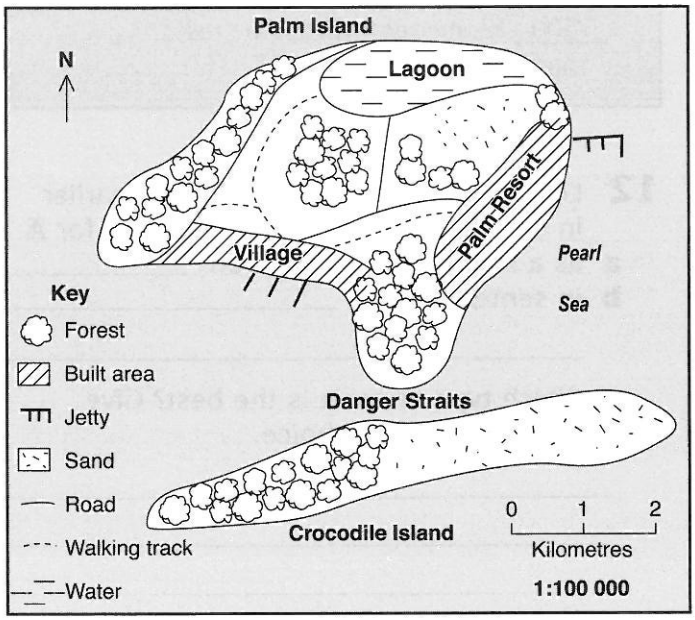
large-scale map shows a lot of detail over a small area

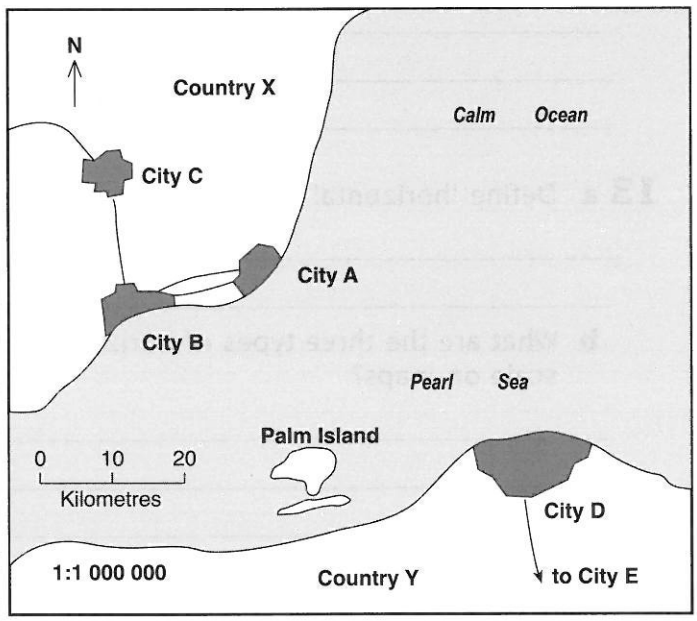
small-scale map shows few details over a large area

ratio is a comparison between two numbers or amounts

representative fraction is a ratio scale

denominator is the bottom number of a fraction





Large scale and small scale

Palm Island is a small island with a local population living in a village near the sea. The Palm Island Resort is a popular tourist destination. The two maps of Palm Island are drawn at different scales – one is at a larger scale than the other.

1 Look at the two different linear scales and complete this table:

Palm Island maps	Sentence scale	Ratio scale (representative fraction or RF)
Top map		
Bottom map		

2 Look at the top map:

a What does 'built area' mean?

_____

b Do you think people live on Crocodile Island? Give evidence.

_____

c Give a general description of the islands, including their size and features.

_____

_____

3 Look at the bottom map:

a How many countries are there? _____

b How many cities are there? _____

c Which country is Palm Island likely to be part of? Why?

_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

4 Compare both maps of Palm Island:

a List 5 things that you can see in the top map but not in the bottom.

_____

_____

b List 5 things that you can see in the bottom map but not in the top.

_____

_____

c Which map would be the most useful for a tourist wanting to walk around Palm Island? Why?

_____

d Which map is at the larger scale? Explain.

_____

_____

5 Use these terms to complete the two paragraphs about different map scales:

- ground
- ratio
- size
- larger
- objects
- map
- small
- $\frac{1}{5\ 000\ 000}$
- representative fraction

To understand the use of the terms 'large scale' and '_____ scale', first think about the ratio or RF (_____ _____) method of showing map scale: The _____ 1:1000 means that the size of _____ on the map is $\frac{1}{1000}$ of their _____ on the ground.

The ratio 1:5 000 000 means that the _____ of objects on the _____ is _____ of their size on the _____

$\frac{1}{1000}$ is a _____ fraction than $\frac{1}{5\ 000\ 000}$ so 1:1000 is the large-scale map.

MAP SCALE RULE

The larger the **denominator**, the smaller the scale. The smaller the denominator, the larger the scale.

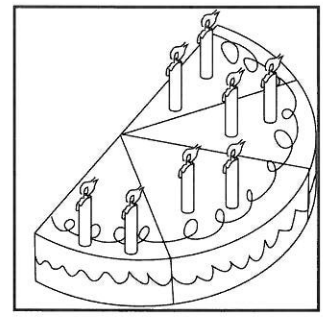
6 Complete the drawings below and answer the questions.

a The left side shows half a cake with 8 candles. This is cut into 4 wedges. Draw one of the wedges in the right box. Make the sides 3 cm long.

b How many candles are there on the wedge? _____

c Which one is the larger-scale drawing?

_____



7 If you redraw the Palm Island 1:100 000 map on the previous page at a scale of 1:50 000:

a What will be the length and width of this map? _____

b Is this a larger scale map or a smaller scale map than the first map? _____

c Redraw the map on a sheet of paper or in your workbook.

8 Reread the information box at the start of this worksheet. Explain the differences in scale that Max and Paula would have seen on their two different journeys.

_____

_____

_____

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

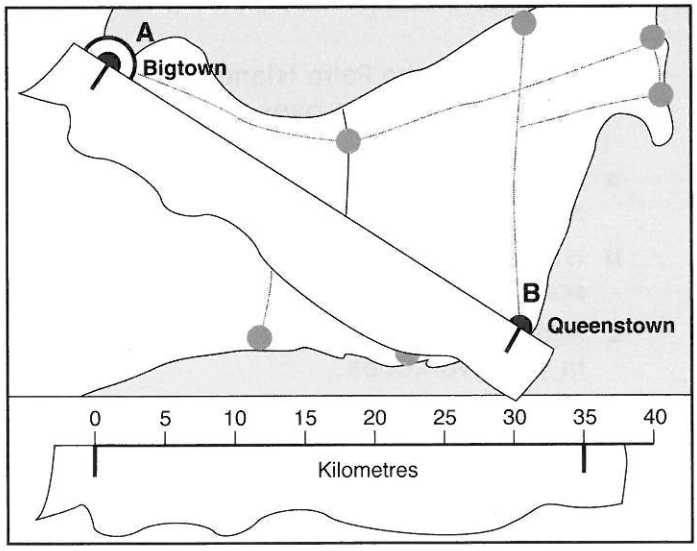
10 Measuring straight-line distance

NOTE TO TEACHER
This worksheet uses the map of Cape Beauty on page 42. Students will need a copy of this map.

You can use the scale of a map to calculate the distance between any places or features on a map. You can work out the straight-line distance between two places. That is the distance 'as the crow flies' or as an aeroplane flies. It is the shortest distance between two points.There are different ways of calculating straight-line distance just as there are different ways of stating the scale.

Using the linear scale

You can use the linear scale to work out the real distance between two points on a map. Use a straight edge of a piece of paper and lay it on the map so the edge of the paper touches both points. Mark the two points on the piece of paper. Place the paper against the linear scale with the first mark aligned with the zero position on the linear scale and the second mark along the scale. Use the second mark to read the real distance off the scale. If the second mark is between numbers on the scale, estimate the reading from the numbers before and after the mark.



1 In the map above what is the distance between Bigtown and Queenstown?

_____

Use the map of Cape Beauty to answer questions 2 to 9.

2 What is the straight-line real distance between the swimming pool and the nearest parking? _____

3 What is the straight-line real distance between the ambulance station and the nearest church? _____

4 Is the straight-line distance between the edge of the fire station and the marina shorter or longer than the straight-line distance between the right edge of the high school block and the railway station?

_____

5 In question 4, what is the difference between the two straight-line distances?

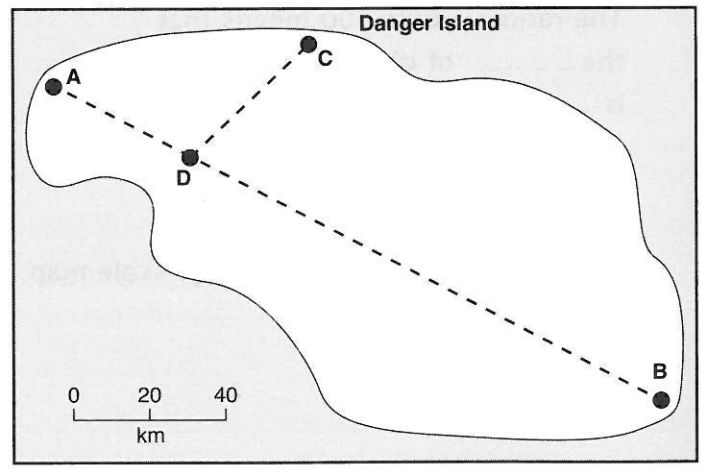
_____

What if the distance between two points on the map is greater than the length of the linear scale? A different approach is needed.

Two approaches are to:

- use the scale multiple times
- use simple multiplication.

An example: two places A and B on Danger Island are 9 centimetres apart and the linear scale shows that 2 centimetres represents 40 kilometres. The table on the next page shows the steps to follow in using the scale multiple times.



© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Step	Purpose	Process	Result
1	Measure and calculate the distance represented by the full length of the scale.	Align the first mark on the zero position on the scale. Make a small mark at the end of the 40-kilometre point. Count the number of 40-kilometre lengths that have been measured so far (label as count x 40 kilometres if you wish).	Now have a running total of the real distance as the count number times 40 kilometres.
2	Decide if the remaining distance is more than the full length of the scale. Yes or No?	If Yes, make the latest mark the starting point for the next measurement and repeat step 1. If No, continue with step 3.	
3	Measure and calculate the distance represented by the remaining distance.	Align the latest mark at the zero point and read off the remaining distance.	Add the remaining distance to the running total to get the final straight-line distance on the ground.

In the example given, the process would be to perform:

- step 1 giving a running total distance of 40 kilometres
- step 2
- step 1 giving a running total of 80 kilometres
- step 2
- step 1 giving a running total of 120 kilometres
- step 2
- step 1 giving a running total distance of 160 kilometres
- step 2
- step 3 giving a final distance of 180 kilometres (the remaining distance of 20 kilometres).

6 Use this approach to find the real distance between:

a the high school and the Cape Beauty sports field _____

b Cape Beauty sports field and the golf clubhouse _____

c the high school and the golf clubhouse.

_____

7 Use the map of Danger Island. What is the straight-line distance from:

a C to D? _____

b D to B? _____

c C to B? _____

This table shows the steps to follow using simple multiplication.

Step	Purpose	Process	Result
1	Calculate the real distance used by a simple measuring unit on the map.	Read off the scale what distance is represented by 1 centimetre on the map. Or calculate such as if 2 centimetres represent 40 kilometres, then dividing by 2 will give one centimetre represents 20 kilometres.	Have a simple-to-use scale of 1 centimetre represents 20 kilometres.
2	Get the distance on the map between the two points in simple measuring units.	Measure the distance between the two points.	Measured distance of 9 centimetres.
3	Calculate the real distance between the two points on the map.	Multiply the result of step 1 by the result of step 2.	180 kilometres

8 Use this approach to find the real distance between:

a the swimming pool and the high school

b the swimming pool and the primary school

c the lookout and the swimming pool.

9 Mobile phone coverage in Cape Beauty is limited to 1.3 kilometres from the tower. The mobile phone tower is at the lookout. Is there mobile phone coverage at:

a the golf course?

b the railway station?

c the resort?

d the sports field?

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

11 Measuring non-linear distance

NOTE TO TEACHER
This worksheet uses the map of Cape Beauty on page 42. Students will need a copy of this map.

You can use the scale of a map to calculate the distance between any places or features on a map. However, it is not always possible to travel from one point to another in a straight line. You have to be able to measure non-linear distances around corners and curves. You can work out how far it is between two points travelling along a road or a river, or even around a coastline.

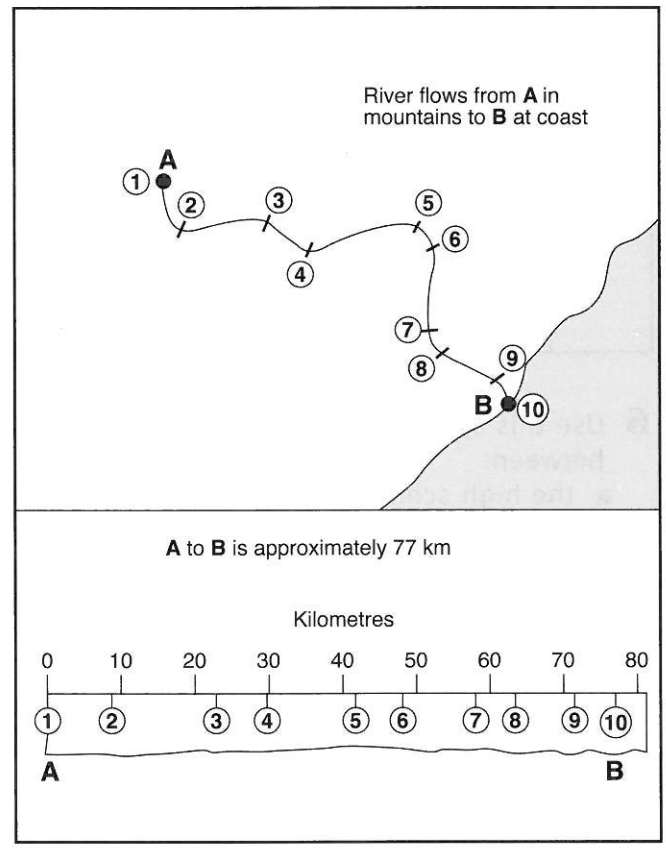
Curved distance

You can measure distance along curved lines such as a river or road, using the straight edge of a piece of paper.

Place a mark on the paper and locate this mark at the start point of the map from which the curved line is to be measured. Align the edge of the paper along a straight portion and make a mark on both map and paper when the edge of the paper leaves the straight portion of the line being measured.

Rotate the paper until another straight portion of the curved line is aligned with the edge of the paper and mark the end of that straight point. Repeat this process until the measurement is completed.

Place the paper against the linear scale with the first mark aligned with the zero position on the linear scale. The last mark will be aligned on the scale with the real distance. Ignore all the in between marks.



1 Why are travelling distances not always in a straight line?

_____

_____

2 On the Cape Beauty map:

a What is the real distance from the post office to Beach Parade if you pass the primary school? _____

b What is the shortest way by car from the sports field to the resort? _____

c What is the real distance? _____

3 Could you use a piece of string to measure distances that are not in a straight line?

_____

4 Write an explanation of how to use a piece of string to calculate the real distance following a curved road between two points on a map.

_____

_____

_____

_____

5 Name two other things that could be used to measure distances around curves.

_____

_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Approximating distance and travel time

As maps are drawn to scale, they often have to leave out much of the detail on curvy bits. We all know that there are many more bends in a river or stream than can ever be shown on a map. We can only approximate the real length.

The length measured along a curve on a map will depend on the scale of the map. Large-scale maps show more objects and features than small-scale maps, so they show more curves.

When distances are very long, such as from the east to the west coast in Australia, they are usually only an approximation when calculated from a map. Again the scale of the map will affect the approximation.

People use distance to help them make other approximations, such as the time that it takes to travel from one place to another. But distance on maps, whether straight-line or curved, does not take into account the flatness or steepness of an area.

6 Use your research skills to find out the length of the coastline of Australia. Obtain your answer from at least three sources. Explain any differences.

_____

_____

_____

7 Use two maps of Australia at two different scales – one at about 1:20 000 000 (1 kilometre to 20 kilometres) and a much smaller scale map, such as 1:80 000 000 – to calculate the approximate distance that a plane would fly between these places:

a Brisbane and Sydney _____

b Sydney and Adelaide _____

c Canberra and Hobart _____

d Alice Springs and Darwin _____

e on a round trip starting from Melbourne to Perth, then to Broome, Darwin, Brisbane and return to Melbourne

_____

_____

f Make a statement about map scale and measuring accurately.

_____

_____

TRAVELLING TIME

Distance measured on a map does not take into consideration the rise and fall of the land. Distances measured by using maps and their scales are flat distances. The distance measured on a map could increase when actually measured on the ground. This must be taken into account when navigating across land.

8 Circle **T** True or **F** False to the following questions about measuring straight-line and curved distances, approximating distance and travelling time.

a Some people think that the distance between two points is always in a straight line because this is the way it appears on maps. **T** **F**

b Where roads and rivers join places the distance between them has to be measured around curves. **T** **F**

c Curves are easier to measure on small-scale maps. **T** **F**

d Distances on large-scale maps are difficult to approximate. **T** **F**

e People often use maps to estimate their travelling time between one place and another. **T** **F**

9 Draw an illustration that shows how the shape of the land affects travelling time.

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

12 Measuring area on maps

NOTE TO TEACHER

This worksheet uses the map of Cape Beauty on page 42. Students will need a copy of this map.

Students will also need a piece of tracing paper at least 10 cm square.

You can use the scale of a map to calculate the area of a feature on a map. A way to do this is to create a series of grid squares over the feature. Use the scale to convert the area of one grid square to the real area. Count the number of grid squares over the feature to calculate the total real area of the feature.

The size of the grid square you choose should be easily related to the scale used on the map.

Creating a grid

Sometimes you might want to calculate the area of a feature on a map. It could be the area of a national park or the area of a lake. You can do this by creating a grid over the area and multiplying the number of grid squares covering the feature by the area represented under each grid square.

One way to do this is to use a piece of tracing paper larger than the area on the map that you want to calculate. On the tracing paper, use your ruler to mark points one centimetre apart along the top and bottom of the paper. Join each pair of marks with a line. Then mark points one centimetre apart along each of the sides and join these. You will now have a grid of one-centimetre squares.

1 Fill in the missing words in the sentences about drawing grids. Select the correct words from this list:

- seen
- grid
- see-through
- purple
- paper
- feature
- map

Tracing _____ is used for the _____ because it is _____.

It can be placed over the feature and the _____ can still be _____.

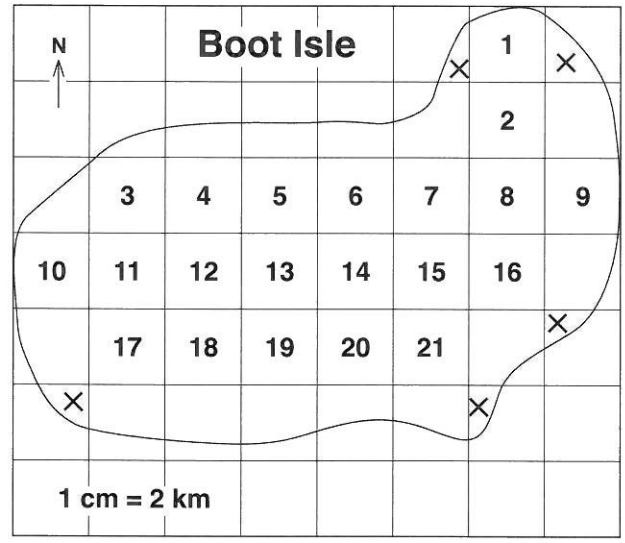
2 There are advantages of drawing the grid on tracing paper rather than directly on the map. Name at least two advantages.

_____

_____

Area of each grid square

Calculate what real area is represented by each square. If the scale is 1 centimetre to 1 kilometre then each square represents (1 x 1) 1 square kilometre. If the scale is 1 centimetre to 0.5 kilometre then each square represents (0.5 x 0.5) 0.25 square kilometres.



3 What is the real area represented by one square centimetre on the map where the scale is:

a 1 cm represents 10 km _____

b 1 cm represents 50 m _____

c 1:5? _____

CALCULATING AREA

The formula to calculate the area of a rectangle is multiply the length by the width. This will give an answer in square units.

A rectangle 4 centimetres long and 5 centimetres wide will have an area of 4 x 5 = 20 square centimetres, or 20 cm^2.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Calculate the area

Place the grid over the map. Count the number of complete grid squares that are in the area to be measured. That is the easy part.

But what do we do with the parts of the feature that are only partly covered by a grid square?

For the incompletely covered squares you could estimate each of them – for example, a half or a quarter or a sixth – and then total these fractions. However an easier way that normally gives a fairly accurate result is to count the number of incomplete squares and use half of that number.

4 If a scale is 1 centimetre equals 10 metres what is the real area represented by:

- **a** 1 square centimetre on the map _____
- **b** 5 square centimetres on the map _____
- **c** 20 cm^2 on the map? _____

5 Use the map of Boot Isle to calculate its area:

- **a** How many complete grid squres are there? _____
- **b** Place a small x in each incomplete grid square. Count the total _____
- **c** Divide the total of incomplete squares by 2 _____
- **d** Add **a** and **c** _____
- **e** How many square kilometres is one grid square? _____
- **f** Multiply **d** by **e** _____
- **g** The area of Boot Isle is _____

6 Draw a grid on tracing paper using 1-centimetre squares. Use the grid to calculate the approximate areas of these places in Cape Beauty.

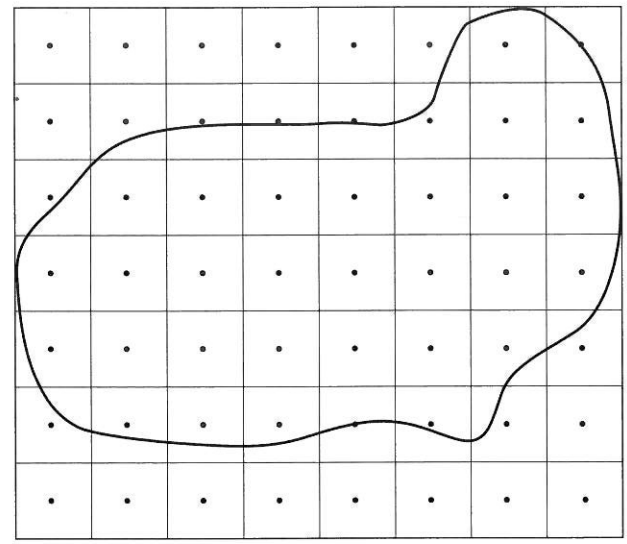
- **a** Cape Beauty sports field _____
- **b** the golf course _____

7 Put on your thinking caps and use the grid to find the area of the reserve that surrounds the Cape Beauty golf course.

_____

Using a grid sheet

Another technique to calculate area is to use a grid sheet. A grid sheet is the same as the grid that you drew with one big difference. It has a dot in the centre of each square. Place the grid sheet over the feature. Count the number of dots within the feature. To get the real area, multiply the real area represented by each square by the number of dots counted.



8 Modify the grid you drew in question 6 by placing a dot in the middle of each one centimetre square.

- **a** Recalculate the area of Cape Beauty sports field _____
- **b** Recalculate the area of the reserve that surrounds the Cape Beauty golf course

_____

- **c** Explain your results.

_____

_____

_____

9 Research other ways of calculating area such as using finer grids or using planimeters.

_____

_____

_____

_____

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

13 Representing height on maps

NOTE TO TEACHER
This worksheet uses the map of Lake Mystery on page 43. Students will need a copy of this map.

DEFINITIONS
2D is the abbreviation of a two-dimensional area having length and width
3D is the abbreviation of a three-dimensional, being 2D + height
cartographers are people who draw maps
relief is the difference in height between places
elevation is the height of the land above sea level
depth is how deep the ocean is at a point

Last week Max Quickdrawer and Paula Mapmaker went on a hot-air balloon flight. The flight left from the park near their school. They took a map with them to help find features that they flew over.

Max and Paula realised that most people think of location only as **2D** or flat. They are not aware of the important third dimension of our surroundings height.

Cartographers use several different ways of showing **3D** shape within the 2D map.

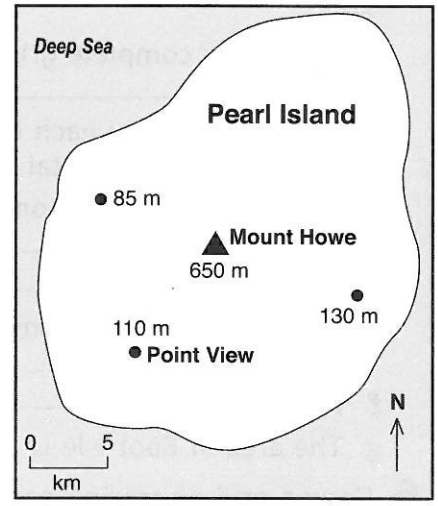
1 Maps are printed on a flat surface or displayed on a computer.

a Write one sentence to explain why most maps show a flat view of an area.

_____

b Why would most users of maps want them to show some things in 3D?

_____



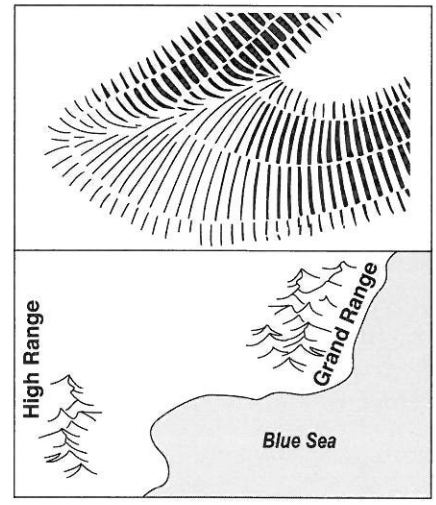
Spot heights

The simplest way to show a vertical measure on a map is to label a point with an **elevation** or depth measurement. The spot is represented by a small point symbol with a number beside it indicating the height or depth. This is the spot height, or depth if in water. Sometimes the point is also named.

Hachures

A hachure is a quick, hand-drawn short line. A series of hachures can be drawn on a map to give the idea of mountains, hills, valleys and rivers as in the diagram at right. They might not give exact heights but a general impression of an area on a large-scale map. They are difficult to draw and take a long time.

On small scale maps of whole countries simple hachures are used to show mountain chains or groups. They are not drawn to scale. They are just like another symbol that is used for mountains and are shown in the key.



2 Spot heights are point symbols.

a What does this mean? _____

b Why do you think that a spot height is often labelled with a name? _____

c Why is a spot height the simplest way of showing height on a map? _____

3 Are hachures more accurate than spot heights in showing elevation? Explain. _____

_____

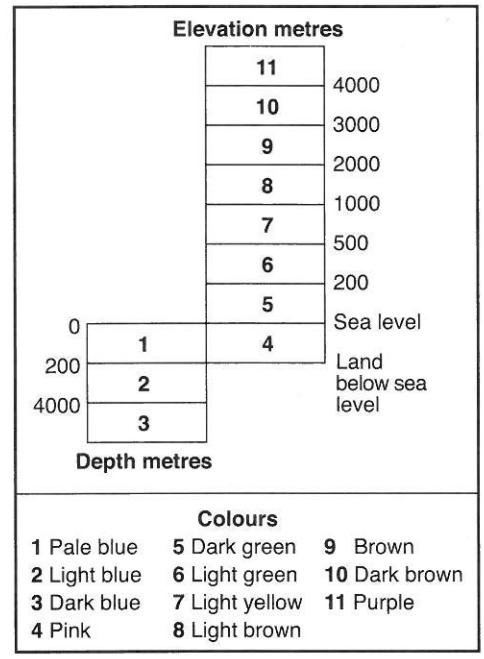
© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Layer colouring

Colours can be used to show heights on a map. A different colour or shade of colour is used for each layer of elevation. The elevation band is a range from one height to another, as shown in the diagram at the right.

Blue is used for water and different shades of blue are used to show the layers of depth in the oceans.

A key on the map shows the height (or depth) layer represented by each colour.



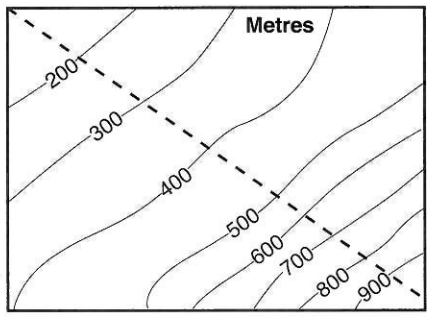
4 The blank boxes to the right can be used for form a key to show height and depth on a map.

- **a** Fill each box with the colour or shade of colour as indicated.
- **b** How many elevation and depth layers are there?
- **c** What is the layer described as 'land below sea level'?
- **d** An area has mountains that range from 3000 to 4300 metres in height. What colour/s is the area?
- **e** Flat land is often found close to the coast. What colour and elevation would it be?
- **f** The lightest blue shade is usually near the coast. What does this mean?

Contour lines

A contour line represents an imaginary line on the ground, above or below sea level. All points on the contour line are at the same elevation or depth.

Contour lines are mostly used on a special type of map called a topographic map. (Topography is the shape of the land.)



5 Examine the pattern of contour lines in the diagram above.

- **a** How much height does each layer increase by? _____
- **b** How did you calculate this? _____

_____

- **c** This difference is called the 'contour interval'. Explain this term. _____

_____

- **d** What is the length in centimetres along the broken line from the left corner to the 500 m contour? _____
- **e** What is the length along the same line from the 500 m to 900 m contour? _____
- **f** Where is the land steeper? Why? _____

Local relief

Local relief is the difference in height between the lowest and highest point of an area. Contour lines give the most accurate reading, although layer colouring can also be used to give a range.

6 **a** Estimate the local relief in the contour diagram on this page. _____

- **b** An area of a map has from layers 6 to 10 in the key at the top. What is the range of local relief?

_____

7 Use the base map of Lake Mystery:

- **a** Name the feature that already has a height indicator and state what type of indicator it is. _____

_____

- **b** Mark these spot heights on your map:
  - 1 cm south-east of airport - 810 m
  - half-way between edge of parkland and airport - 850 m
  - middle of parkland - 1300 m
  - middle of tent campground - 930 m
  - top parking area off Mountain Hwy - 1150 m
  - 2 cm north-west of the chalet - 1220 m
- **c** Just south of the airport is the lowest elevation of the area and Mt Whitehorn is the highest. What is the local relief of the Lake Mystery area? _____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

14 Finding things on maps

A grid is a series of equally spaced lines running across and down a map. It is used to quickly locate places on the map. Letters or numbers are placed along the sides of the grid to give the grid reference for a place.

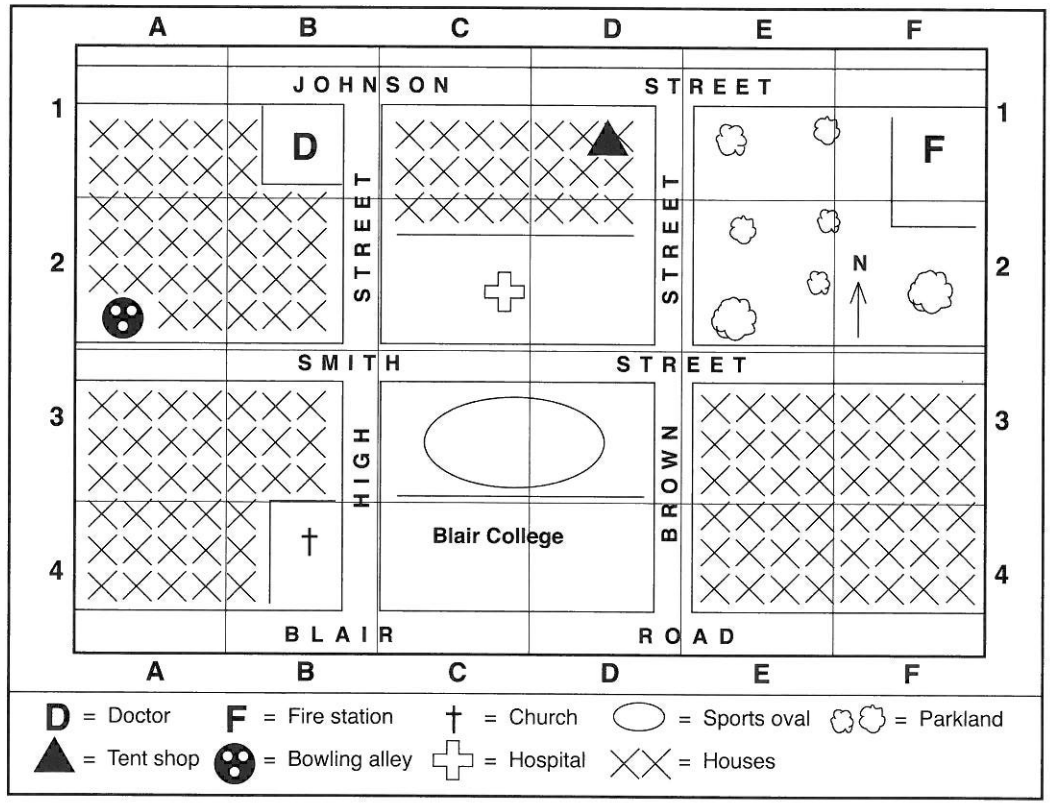
Last week Max Quickdrawer and Paula Mapmaker went on a hot-air balloon flight. The flight left from the park near their school. This is the map that Paula drew. It shows the whole area around her school. To make it easy to find some of the places on the map, Paula has set up a grid.

DEFINITIONS

alphanumeric is a combination of letters and numbers

grid coordinate is the reference for the an area on a map covered with a grid system

area reference is a four-figure number made up of a two-figure eastings reading and a two-figure northings reading used to define a location within an area



Paula has drawn lines horizontally and vertically over her map to divide it into grid squares. On the top of the map she has placed the first 6 letters of the alphabet in the middle of the grid lines. In the same way, on the side she has put the numbers 1 to 4.

This type of grid is called an **alphanumeric** grid. It can be used to easily find features on a map.

Paula told Max that there is a sports oval at grid coordinate C3. So Max put one finger on the C and another finger on the 3. He moved the C finger down and the 3 finger across until they met to find the grid location C3. He quickly found the oval by just looking in that square bounded by the **grid coordinates**.

1 List one or two features that appear at these grid coordinates:

a B4 _____

b A2 _____

c D1 _____

2 Give the grid coordinates for:

a park _____

b doctor _____

c fire station _____

d hospital _____

e corner of Brown Street and Blair Road

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

3 On a separate sheet draw a map of your classroom using symbols for objects such as desks, tables, teacher's table, blackboard, notice board, windows, door, computers and pot plants.

a Add an alphanumeric grid in black with at least 6 squares running across and down. Along the top of your map, label your squares starting with A for the first square, B for second square, and so on. Down the side of your map label your squares the numbers starting with 1 from the top.

b Explain why the grid is called an alphanumeric grid.

_____

_____

c List 6 things on the map and their alphanumeric reference (grid coordinates).

_____ _____

_____ _____

_____ _____

d Add another alphanumeric grid to your map in red using only A and B horizontally and 1 and 2 vertically.

e List the same 6 things from question (c) and their references using the grid you drew in red.

_____ _____

_____ _____

_____ _____

f Is it easier to use the black grid or the red grid to locate features? Explain why.

_____

_____

4 If Paula had not drawn the alphanumeric grid it would be more difficult to find the church on the map. Without using grid references, write instructions to tell Max where to find the church on the map.

_____

_____

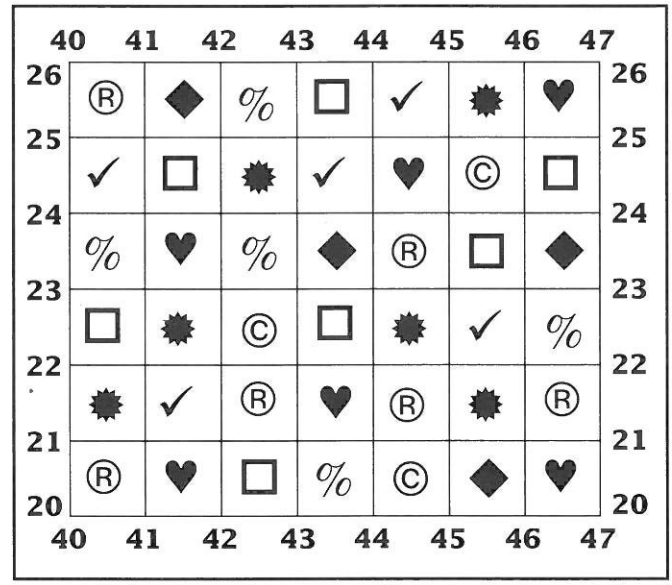
_____

_____

_____

_____

Area reference



This diagram uses another form of grid system. It is made up entirely of numerals located on each grid line. The numbers across the page are called *eastings*. They increase in value as you move east, to the right across the page. The numbers that measure up and down the page are called *northings*. They increase in value as you move north up the page.

An **area reference** for a point joins the two numbers from the eastings reference with the two numbers from the northings reference to make the four-figure area reference.

The ® symbol is located in the square with an eastings of 44 and a northings of 21 – its area reference is 4421.

5 Draw the symbol at these area references:

a 4124 _____ **c** 4325 _____

b 4425 _____ **d** 4522 _____

6 Trace the map of the classroom you did in question 3 with the black grid lines but without the alphanumeric grid references (or redraw it if you like).

a Add *eastings* on each grid line across the top of your map. Number each grid line as in the diagram above. Start at 50 and increase by 1 as you move across the map.
Add *northings* on each grid line along the side of your map. Start at 20 for the grid line at the bottom of the map and increase by 1 as you move up the map.

b List 6 objects on your map and their area reference.

_____ _____

_____ _____

_____ _____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

15 From area references to grid references

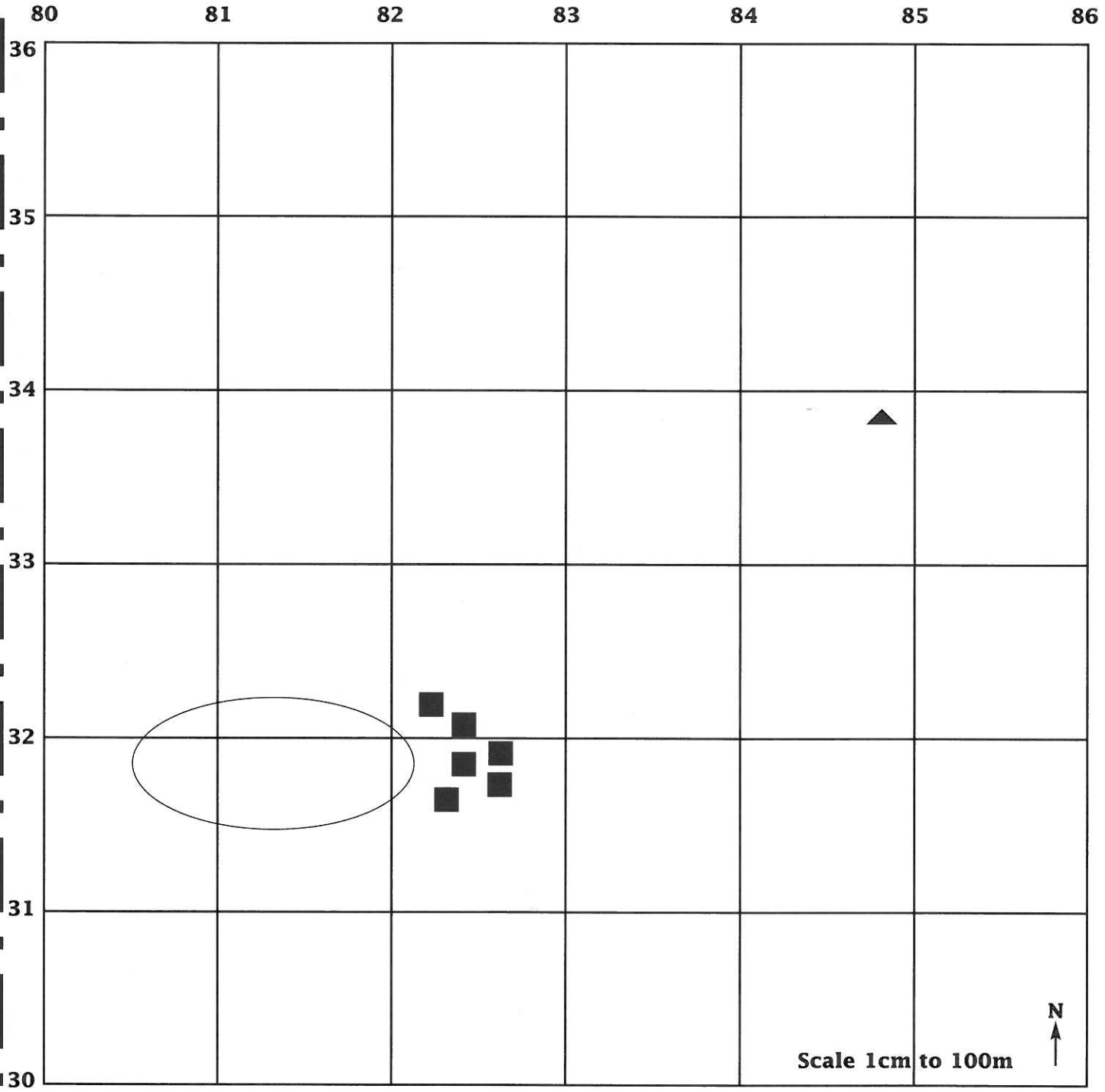
NOTE TO TEACHER

This worksheet uses the map of Lake Mystery on page 43. Students will need a copy of this map.

DEFINITIONS

area reference is a four-figure number made up of a two-figure eastings reading and a two-figure northings reading used to define a location within an area

grid reference is a six-figure number made up of a three-figure eastings reading and a three-figure northings reading



1 Add the correct labels to the bottom and right grid lines. State why you chose them.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

2 What type of system is shown on the map on the previous page? What type of reference would you use to locate something on it?

_____

_____

3 There are three symbols used on the map representing a lake, some houses near the lake and a lookout.

a Give the **area reference** of:
the lookout _____
the group of houses _____
the lake _____

b What did you notice when you had to give the area references for the houses and lake?

_____

_____

_____

_____

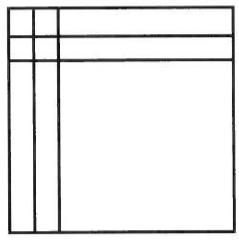
c If you did not use the middle of each of these features, restate their area references:
houses _____
lake _____

4 Each grid square measures 3 cm by 3 cm.

a How many millimetres is this
_____ mm by _____ mm.

b You have to divide the square into tenths.
How wide is each division? _____ mm

c Practise dividing a 3 cm square into tenths by finishing this:



5 a Use this measurement to divide the whole grid square containing the lookout into tenths east and north. Draw in the 9 lines at even intervals.

b The first eastings reading is 84 + 1 (the first tenth) which is 841 and the last is 84 + 9 = 849. What are the first and last northings reading in the square?
_____ and _____

c What are the middle eastings and northings reading in the lookout square?
_____ and _____

Grid references

Grid references are used to more accurately pinpoint a feature than an area reference. This is done in your mind by subdividing each grid square into tenths. These subdivisions do not appear on the map.

This allows you to estimate in your mind a three-figure reading for the eastings and another three-figure reading for the northings just as you did in question 4, but without having the drawn subdivision lines.

By combining both three-figure readings you get a six-figure reference, which is known as a grid reference. Therefore the lookout is at grid reference 847338.

6 Give the grid references for the middle house _____ and the middle of the lake _____.

7 Create symbols for the following features and draw them in on the map:

a a group of 10 houses centred on 805355

b a road connecting the two groups of houses

c a straight walking track connecting the lake houses to the lookout

d a forest on the west side of the lake that covers the rest of 8031 and 8032

e a river flowing south-west from 856360 to 818323

f How long is the walking track? _____

8 Add a key above the map.

9 Create a grid system of 2 cm squares over the Lake Mystery map, starting with 53 as the first eastings reading and 62 the last, and 25 as the first northings reading and 33 the last. Give the grid references for:

a the police station _____

b Lake Mystery Chalet _____

c the airport _____

d the warning beacon _____

10 Name four other features on the map and give their grid references.

_____

_____

_____

_____

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

16 Mapping a tribal group area

A pictorial map is a special type of map that contains small graphics and line drawings to replace the usual mapping symbols. The pictures are not drawn to the same scale as the map area. The pictures are at a larger scale in relation to the map scale.

Pictorial maps are often drawn by people who do not have trained mapping skills but who want to use a map to show the location of features over an area. Many of the first maps ever drawn by humans were pictorial as the drawers scaled down what they could see from a high point or what they saw as the walked from place to place.

1 Imagine that you are part of a small tribal community of 50–60 people living near a river. The main way that your group gets food is by hunting wild animals and birds, fishing and gathering wild fruits, nuts and honey. The group also has a small garden plot next to their huts. Members of the group often explore up to three days walk away from the settlement to locate food.

Your community decides it wants to make a map on a large flat rock to help them find valuable food sources after they have been discovered.

Complete this table to show the things in the surroundings that would be important to put on the map. In the last column, draw a small picture of how you would show the ones you ticked on the map – just like the first one, which has been done for you.

If you have access to the Internet, this website will give you lots of ideas for pictures:
<http://www.enchantedlearning.com/coloring/>

Feature and location	Important	Not important	Reason	Picture
Part of river where there are large juicy fish, 1 day's walk down river	✓		Good eating and feeds a lot of people	
Part of river shore where there are small poisonous frogs, close to the huts				
Trees with fresh honeycombs, $^1/_2$ day's walk to the left of huts				
Part of river where there are lots of small fish, close to the settlement				
Forest with many different fruit and nut bushes, also birds, $^1/_2$ day's walk up river				
Lake where animals drink such as deer, buffalo and large birds, 2 day's walk towards afternoon sun				
Field where wild turkeys nest, 3 day's walk towards afternoon sun				

(This table is continued over the page.)

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

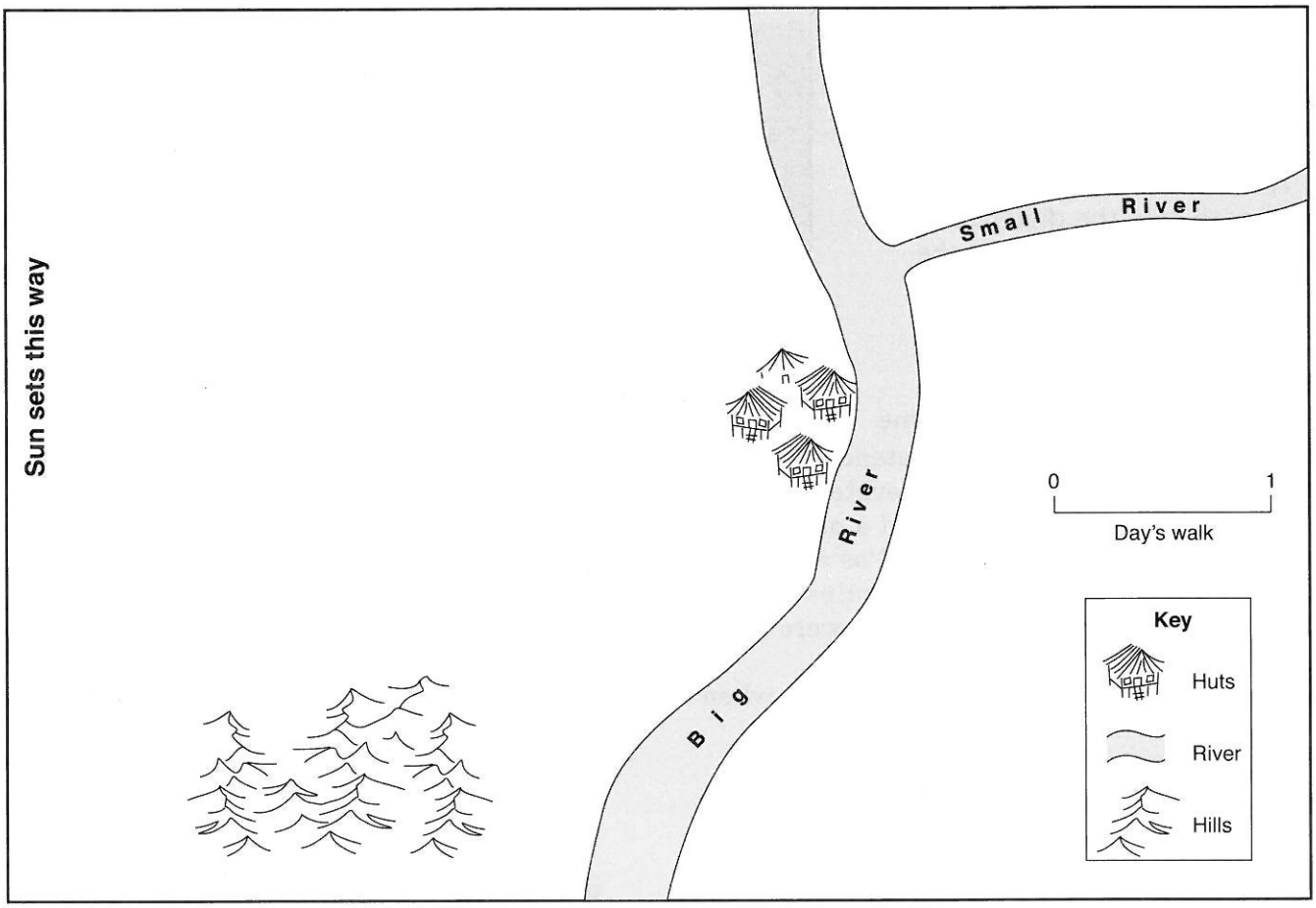
Feature and location	Important	Not important	Reason	Picture
Very high hills where mountain goats live, 2 day's walk down river then 1 day's walk towards the afternoon sun				
A section of the river where crocodiles live and hunt, where big river meets small river				

2 Which things did you ✓ 'Not important' ? Explain why you have left out any feature.

_____

_____

3 The map below shows the location of the tribal village in relation to the river and hills. Use the instructions in the table and your pictures to complete a pictorial map of your tribe's area.



Modern pictorial maps

Modern pictorial maps are used as a way of attracting users to a special theme. Individual graphics are drawn accurately and are easily recognisable.

The pictorial symbols are placed in a general area of the landscape but the shape and scale are often not exactly to scale.

4 Group activity

Form into small groups. Decide on a special theme at a place that could be mapped. Draw a pictorial map to illustrate the theme. (Hint: you may need to copy or trace a base map of the area then add your pictures.) Some possible themes are the zoo, local tourist attractions, a historical town or part of your local area, or a recreation park. Add a title, north pointer, scale and key to your map.

© Blake Education 2006

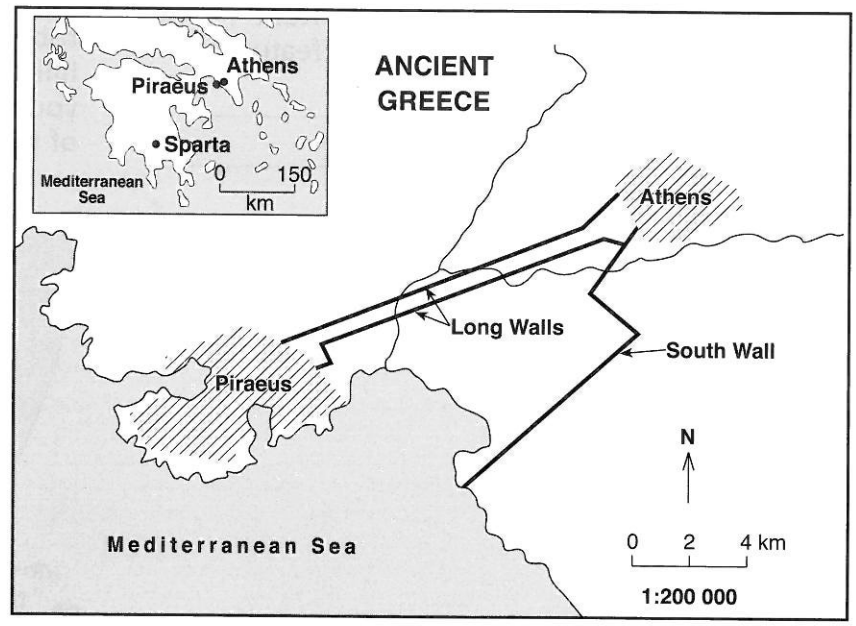
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

17 Mapping ancient Athens

The city of Athens 2500 years ago looked very different to what it does today. However many of the old buildings have been preserved and are now famous tourist sites.

Ancient Athens was a walled city, meaning that there was a huge wall built around it to help protect it from enemy armies. The wall continued to the harbour port of Piraeus, which was one of Athens' main links to the rest of the ancient world.

The main map at the right shows the relationship of ancient Athens to Piraeus. The small map is called an inset map because it is placed inside the border of the main map.



1 What does the inset map show? Is it a smaller- or larger-scale map than the main map? Why?

2 Describe the relationship of Piraeus to the ancient city of Athens, including the location of each city, the distance between them and what the Long Walls show.

The map on the next page is a much more detailed map of the ancient walled city of Athens. One main feature is the Acropolis plateau – a flat-topped rocky hill – that is elevated from the surrounding landscape. It had a natural spring water supply and was used as a fortress. The plateau contains the ruins of many buildings. The most dominant were the Parthenon temple and the Theatre of Dionysus, which could seat 17 000 people when it was built more than 2450 years ago.

Use this map and the one over the page to answer the remainder of the questions and to mark on various geographic features.

3 The walled city of ancient Athens contains some natural features such as hills and rivers, and human features such as buildings and roads:

- **a** What were the longest dimensions of the city from east to west, and north to south?
- **b** Calculate the approximate area of the walled city. Explain how you did this.
- **c** Calculate the approximate length of the wall. Explain how you did this. (Hint: do the calculation twice to test your answer.)

4 There is a road leading from the south-west of the main wall.

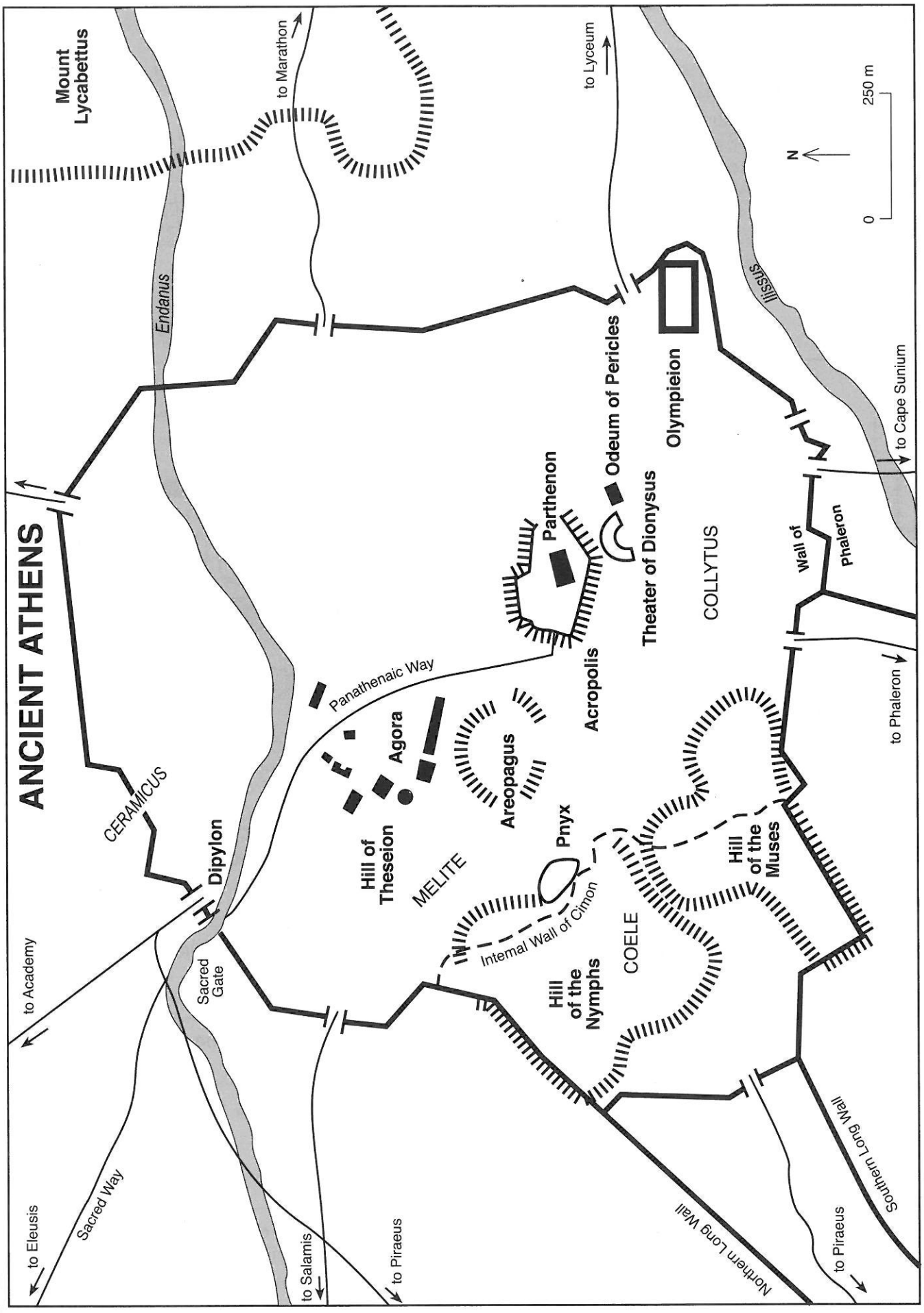
- **a** Where is it leading?
- **b** What does it run between and why?
- **c** Write in a label after the place name to show the approximate distance to it.
- **d** How can you tell where the other roads are? How many are there?

5 There are five main hills on the large map, four inside the walled city and one outside.

- **a** What method of showing height is used for these hills?
- **b** Add a spot height of 187 m to the left of the Parthenon on the Acropolis.
- **c** The Acropolis was the highest point of the walled city. The elevation of Athens is 107 m. What was the local relief?
- **d** Why do you think the internal wall of Cimon might have been built where it is?

6 Add colour to your map of the walled city and a key – pink for roads, red for buildings, blue for rivers and black for the walls.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1



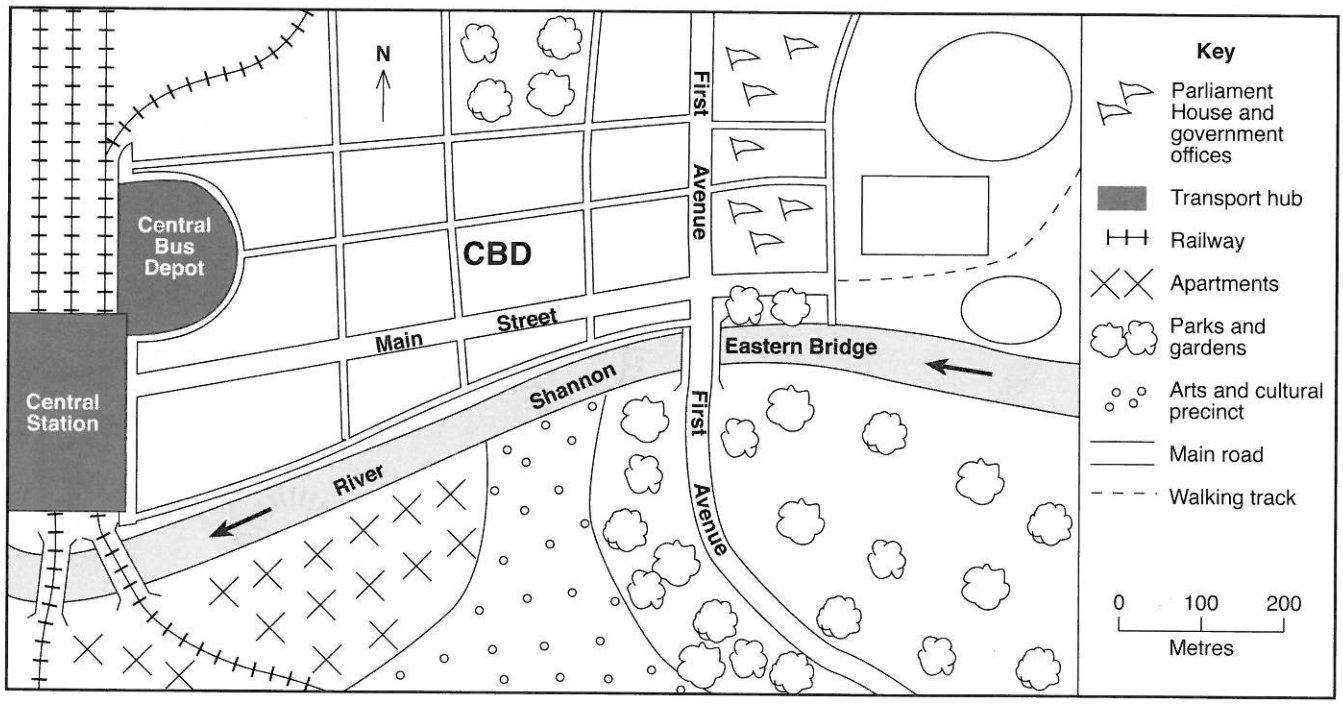
© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

18 Different views of a city

Large modern cities often have a number of distinct **land-use** zones. Although some of these zones can be identified by looking at photographs often a map is a better way of showing this information.

Max Quickdrawer and Paula Mapmaker recently took a balloon flight over their capital city. Max drew some views of the city and Paula created this map, which shows some land-use zones in a large city.



Some of the land-use zones shown are:

- Central Business District (CBD)
- Arts and Cultural precinct
- Sporting Precinct
- Transport Hub
- Parks and Gardens
- Government and Administration area
- High-rise apartments.

If you look at the key to the map you can see how the different areas are identified by the use of different shadings and symbols.

DEFINITIONS

land use is the main type of activity that happens in an area, such as farming, industry, recreation, business

sketch map is a quickly drawn map showing some main features

1 What shading is used to identify the transport hub?

_____

2 Name 2 features identified by labels in the transport hub.

_____

3 How can you identify the sporting precinct?

_____

4 What land-use zones are near the sporting precinct?

_____

_____

5 If you paddled a canoe down the river starting at Eastern Bridge, name the land-use zones you would pass.

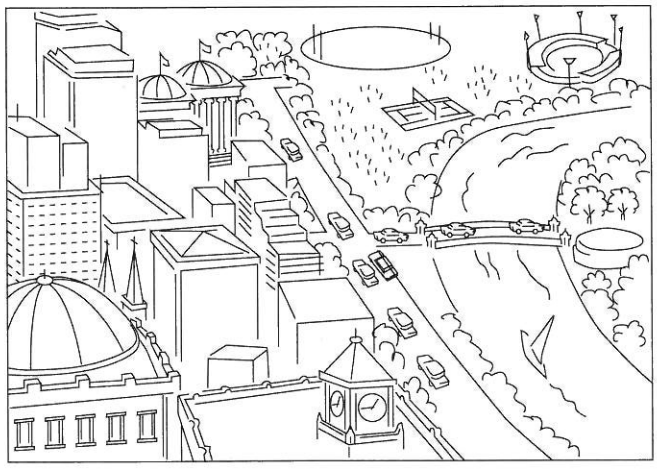
_____

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

As Max was standing on the south bank of River Shannon he did a sketch looking towards the city. At home he used it to draw a ground view on his computer. Look carefully and see if you can identify any of the land-use zones.

6 For each zone you can identify:
- **a** name the zone
- **b** list some of features that you can recognise.

7 On a separate sheet, make a ground-view drawing of what Max would probably see if he was standing on the north bank of the river 200 metres west of the bridge. (Hint: identify the zone/s then imagine the types of building or land that would suit the zone/s and draw this from side on.)



This is Max's oblique-view drawing of the city from out of the window near the top of a very tall building.

8 List 5 or more things that you can see in the oblique drawing that you cannot see in the ground view.

9 List 3 or more things you can see in the Main Street drawing that you cannot see in this drawing.

How well do you know the land-use zones around your school? Of course, there is a school or educational zone. There may be a residential or housing zone. There may be a factory or industrial zone. There may be other land-use zones. There will probably be some mixed-use zones, too.

10 Create a list of the land-use zones around your school. Name and describe them.

11 Draw a **sketch map** showing the land-use zones around your school. Just show the zones and some of the main roads. Use different colours as the symbols for each zone or area. Show these in your key. Give your map an approximate scale.

12 Go back to Paula's map of the city. Choose different colours for each land-use zone. Apply the colours to the map and the key.

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

19 Multiple choice test

For each of the following questions circle **a, b, c** or **d** for the **best** answer. If you change your mind, put a cross through the circle, and circle another choice.

1 There are three main ways of viewing features on the earth:
- **a** ground, horizontal and vertical
- **b** oblique, plan and vertical
- **c** side-on, horizontal and vertical
- **d** ground, oblique and plan.

2 A ground-view of the surroundings is drawn:
- **a** by lying flat on the ground
- **b** from the top of a mountain looking down at the ground
- **c** as a side-on view from normal height level
- **d** by flying over an area.

3 A map is:
- **a** a type of graph
- **b** a type of drawing
- **c** a precise plan of a place
- **d** all of the above.

4 Most maps show this view:
- **a** a bird's-eye view
- **b** a plan view
- **c** a vertical aerial view
- **d** all of the above.

5 Symbols are used on a map:
- **a** to make it look colourful
- **b** to identify features and objects
- **c** to help find bearings
- **d** to help measure distances.

6 Another name for the key to symbols is:
- **a** legend
- **b** scale
- **c** compass
- **d** height.

7 Colour is used on maps to:
- **a** make them easier to read
- **b** represent different features and areas
- **c** show different height levels
- **d** all of the above.

8 The three groups of symbols on maps are:
- **a** linear, point and distance
- **b** point, area and scale
- **c** linear, point and area
- **d** none of the above.

9 The north pointer on a map shows:
- **a** how to get around
- **b** the orientation of the map
- **c** the distance between places
- **d** the compass rose.

10 If you travel due east from a point, then go south the same distance, what direction would you need to travel to get back to your starting point?
- **a** north-east
- **b** south-east
- **c** west
- **d** north-west.

11 A compass bearing:
- **a** is the angle between north and the direction of an object
- **b** is not expressed in degrees
- **c** can be greater than 360°
- **d** is always less than 90°.

12 If you walk 2 km at a bearing of 90°, then 4 km at 225°, then 4 km at 45°, how far are you from your starting point?
- **a** 4 km
- **b** 3.3 km
- **c** 0 km
- **d** 2 km.

13 A map is a representation of an area of the earth:
- **a** at a smaller scale than reality
- **b** at a larger scale than reality
- **c** at the same scale as reality
- **d** all of the above.

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

14 A 10-kilometre road on a 1:100 000 map measures:
- **a** 1 metre
- **b** 10 centimetres
- **c** 1 centimetre
- **d** 100 centimetres.

15 A map of a local area drawn the same size as a map of Australia is at:
- **a** a smaller scale
- **b** the same scale
- **c** a larger scale
- **d** cannot tell.

16 Scale on maps can be expressed as:
- **a** a sentence
- **b** linear
- **c** a ratio or RF
- **d** all of the above.

17 The straight-line distance between two places on a map is 6 cm. The linear scale shows that 5 cm represents 100 km. What is the real distance between them?
- **a** 10 cm
- **b** 120 km
- **c** 1200 km
- **d** 6 km.

18 To calculate the area of a feature on a map you need to know the:
- **a** the scale
- **b** the height
- **c** where north is
- **d** all of the above.

19 Representing height on maps:
- **a** helps in calculating distance
- **b** gives a three-dimensional perspective
- **c** locates a place
- **d** helps mountain climbers.

20 On a map, contour lines join places of:
- **a** the same population
- **b** equal temperature
- **c** equal annual rainfall
- **d** equal height above and below sea level.

21 Street directory maps normally use:
- **a** a four-figure area reference
- **b** a numeric grid system
- **c** an alphanumeric grid system
- **d** none of the above.

22 A grid reference is more accurate than an area reference because:
- **a** it is made up of eastings and northings
- **b** it is six figures
- **c** it is alphanumeric
- **d** it pinpoints a feature more exactly.

23 A lookout at grid reference 815237 is further north than a picnic area at:
- **a** 830287
- **b** 859212
- **c** 815238
- **d** 899299.

24 A grid system with squares of 4 cm each over a map:
- **a** is less precise than squares of 2 cm
- **b** is more precise than squares of 2 cm
- **c** is just as precise as squares of 2 cm
- **d** cannot tell.

25 Pictorial maps are often used to:
- **a** give accurate location
- **b** give accurate scale
- **c** emphasise a particular theme
- **d** all of the above.

26 Land-use zones on a map show:
- **a** how tall the buildings are
- **b** the main types of activities in different areas
- **c** the weather pattern
- **d** the distance from the sea.

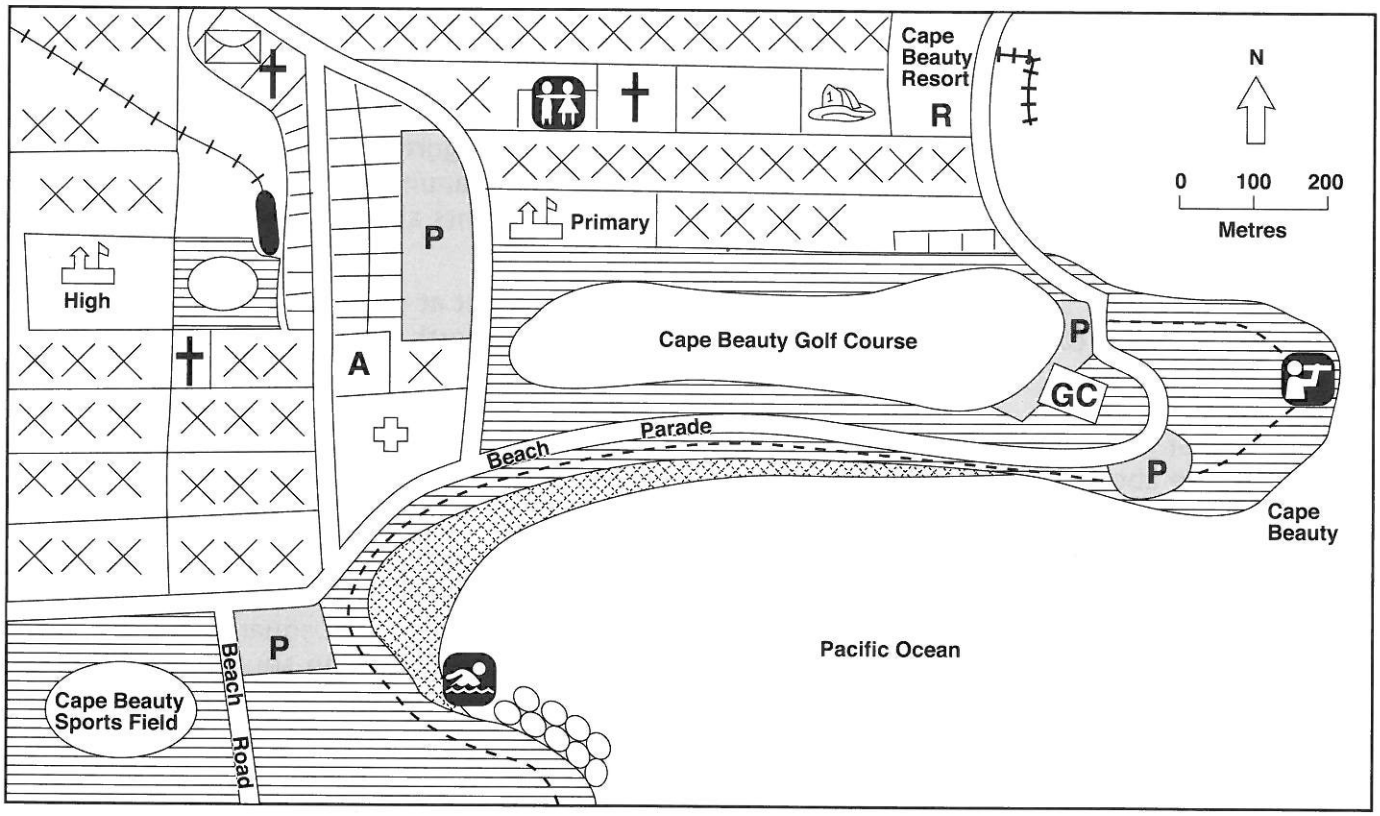
27 Maps are useful because they:
- **a** help users locate places and objects
- **b** can be used to calculate distances between places
- **c** accurately represent a part of the earth's surface
- **d** all of the above.

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

20 Base maps

Cape Beauty



Fire Station	Lookout	Minor road	Parkland	Cliff
Hospital	Swimming pool	Walking track	Parking	Beach
A Ambulance	Post office	Railway	Shops	R Resort
GC Golf clubhouse	Kindergarten	Station	Houses	Sports oval
School	Main road	Marina	Church	

© Blake Education 2006
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

© Blake Education 2006

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Selected answers

1 Different views

1 You can see many things such as car, garage, side fence, garden, steps, letterbox, windows, TV antenna, and attic ...

2 You can see many things including the roof of the house, trampoline, vegetable garden, side fence, table and chairs...

3 Four things are trampoline, back fence, garage roof, and swimming pool. There are several others things also.

4 Some of the things are the garage, table and chairs, swimming pool, water tank and trampoline. There are many other things.

5 Five things are trampoline, back fence, garage roof, vegetable patch, and swimming pool. There are others.

6 Five things are clothesline, water tank, trees between the house and garage, back steps and the path to the clothesline.

7 Birds eye view is what a bird would see as it looked down from directly above. A plan view is the view from directly above. So they are the same view.

9

Front door	☑ ground	☐ oblique	☐ plan
Trampoline	☐ ground	☑ oblique	☑ plan
Vegie patch	☐ ground	☑ oblique	☑ plan
House number	☑ ground	☐ oblique	☐ plan
Size of house	☐ ground	☐ oblique	☑ plan
Backyard	☐ ground	☐ oblique	☑ plan
Front fence	☑ ground	☐ oblique	☐ plan

2 Mapping a plan view

2 plan – a bird's-eye view from directly above

4 Ashcroft Beach

5 unnecessary, redundant

6 houses

7 Ⓢ

9 top

10 Answers include beach, car park, parkland, surf club, Deep Creek, safe swimming area and bridge.

11 Yacht, boat, and cars do not appear on map. Trees, houses and sea etc are represented on the map but shown in the drawing.

12 The names of the roads, creek and ocean are not in the drawing. The drawing does not have a scale, a key or a North pointer.

3 Map symbols and key

1 Ambulance station or houses

2 Lookout

4 There are 2 shopping centres – a large one near the railway and a small one north of the golf course.

7 Crosses are used to mark the housing areas, as a picture symbol for houses would make the map too messy.

4 Different types of map symbols

1

c Whitehorn National Park, Bow River, railway station, railway, hostel, developed area, visitor centre and medical clinic.

3 Some linear symbols are track (no bicycles and bicycles permitted), river, major road, other – sealed road, and railway.

5 Some point symbols are buildings, caravan park, picnic area, and parking

7 Some area symbols are developed area, parkland, lake.

10

Linear	Point	Area
Road Tunnel	Police Station	Forest
Canal	Bridge	Fishing area
	Drinking water	
	Picnic area	
	Picnic shelter	

11 Fishing area, forest.(depending on the map scale)

5 Which way to go

1 Answers could include doctor, tent shop, fire station, church, bowling alley, hospital, sports oval, houses and parkland.

2 North, East, South, West

5

a West **b** South

6

a West **b** South **c** West **d** East
e Hospital, tent shop and houses **f** North-east

7 North-west

8

a South-west
b church, houses, intersection of Blair Road and High Street.
c fire station

9 The compass needle points north. The base has not been rotated so the N on the base lines up with the north-pointing magnetic needle.

6 Getting your bearings

1

a south-west **b** south-east **c** north-west

3

a SSE **b** WNW **c** NNW

4 South-east

5

a 270° **d** 112.5°
b 135° **e** 247.5°
c 315° **f** 337.5°

6 Roads and railways are built and can be followed easily. There is nothing built in the air or sea, so precise directions are needed.

7

Direction	Bearings
North	0°
North-east	45°
East	90°
South-east	135°
South	180°
South-west	225°
West	270°
North west	315°

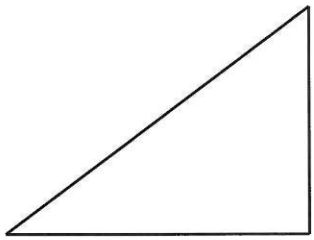
© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

8

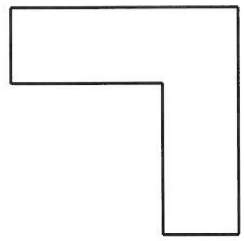
Bearing	Distance	Length of line on map	Direction
90°	4 km	4 cm	east
180°	1 km	1 cm	south
90°	1 km	1 cm	east
180°	1 km	1 cm	south
270°	5 km	5 cm	west
0°	2 km	2 cm	north

9

a



b



c Yes

7 The size of things on maps

1

Calculation

1 cm = 50 cm

3 cm = 3 x 50 = 150 cm

There are 100 cm in 1m.

The animal is 1.5 m long

2 Many answers are possible such as calf, tiger or snake.

3

a 2 cm
b 100 cm or 1 metre
c 3 cm
d 150 cm or 1,5 m
e 2 cm
f 100 cm or 1 metre

6 15 km or 15 000 m

7 35 km or 35 000 m

8 Types of map scale

1

a 1 centimetre on the map represents 1 kilometre on the ground.
b 1 centimetre on the map represents 100 kilometres on the ground.
c 1 centimetre on the map represents 1 kilometre on the ground.

2

a 2 kilometres (km) **b** 5 km **c** 10 km

5

a 2 centimetres (cm) **b** 0.1 cm or 1 mm
c 4 cm **d** 0.001 cm – impossible to show

6 1:500

7 1:200 000

8 100 000 cm or 1000 m or 1 km

9 100 000 glonks

10

a 30 000 cm or 300 m
b 85 000 cm or 850 m
c 5 000 cm or 50m

11

Ratio scale	Distance on the map	Distance on the ground
1:100	4 cm	400 cm or 4m
1:250 000	5 cm	1 250 000 cm or 12.5 km
1:10 000	8 cm	80 000 cm or 800 m
1:10 000	3 cm	30 000 cm or 300 m
1:10 000	15 cm	1.5 km

12

a 1:100 000
b 1 cm on the map represents 1 km on the ground
c Different people find different scales easier to understand depending on their personal skills. People who like maths may prefer ratio scale, while those not good at maths may prefer sentence scale. Visually skilled people may prefer linear scale. An interesting point is that on a small-scale map, ratio scale will give more accurate results than linear scale due to projection distortion.

13 Horizontal scale is the measurement on the map that represents the real distance on the ground. The three types of scale are linear (or bar) scale, ratio (or RF) scale and sentence scale (also referred to as a statement).

9 Maps at different scales

1 Top map – sentence scale is 1 cm on the map represents 1 km on the ground and ratio scale is 1:100 000
Bottom map – sentence scale is 1 cm on the map represents 10 km on the ground and ratio scale is 1:1 000 000

2

a An area where there are built houses, factories, offices and other things made by humans.
b There are probably no or very few people living on Crocodile Island. There is no built area. The western part of the island is all forest and the eastern part is all sand.
c Palm Island is 6 km wide east to west and 4 km long north to south. It is slightly horseshoe shape with a large lagoon in the north and forested areas along the west and in the south-east. Local people live along the southern shoreline and there is a resort along much of the east coast. Crocodile Island is long and narrow, 7.5 km by 1 km. The western half is covered with forest and the east is sand. There is no evidence of people living there.

3

a 2 **b** 4
c Pearl Island is probably part of Country Y as it is less than 10 km from the coast of Country Y and a much greater distance from Country X.

4

a Some of the things are: the built area, resort, lagoon, jetty, walking track, forest, sand and the names for Crocodile Island and Danger Straits.
b Some of the things are Country X, Country Y, Cities A, B, C and D and Calm Ocean.
c The top map because it shows much more detail.
d The top map is a larger-scale map. Large-scale maps show small areas on the ground where the objects are relatively large on the map. Small-scale maps show large areas on the ground and the objects are small on the map.

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

5 To understand the use of the terms 'large scale' and 'small scale' first think about the ratio or RF (representative fraction) method of showing map scale: The ratio 1:1000 means that the size of objects on the map is

$\frac{1}{1000}$ of their size on the ground.

The ratio 1:5 000 000 means that the size of objects on the map is

$\frac{1}{5\ 000\ 000}$ of their size on the ground.

$\frac{1}{1000}$ is a larger fraction than $\frac{1}{5\ 000\ 000}$,

so 1:1000 is the large-scale map.

6

b Two

c The single wedge because it shows a smaller piece in the same area as the half cake.

7

a The current map border box is 9 cm x 8 cm. At a scale of 1:50 000 the border box would be twice this – 18 cm x 16 cm and Palm and Crocodile islands drawn twice the size.

b Larger scale

8 From the hot air balloon Max and Paula would see a smaller part of the earth's surface in more detail than from the plane. The hot air balloon view is at a larger scale.

10 Measuring straight-line distance

1 35 km.

2 The distance on the map is 2 cm and the scale is 1 cm = 100 m so the real distance is 200 m.

3 The distance on the map is 3 cm and the scale is 1 cm = 100 m so the real distance is 300 m.

4 The straight-line distance on the map between the fire station and the marina is 1.5 cm. The distance between the high school symbol and the railway station is 1 cm. So the first distance is 0.5 cm more.

5 As the scale is 1 cm = 100 m, the real extra distance is 50 m.

6 Each 2 cm on the map represents 200 m on the ground.

a The distance is 2 full measures of the scale or 4 cm, so the real-life distance is 200 + 200 which is 400 m.

b The distance is 6 full measures of the scale or 12 cm, so the real-life distance is 200 + 200 +200 + 200 + 200 + 200 which is 1200 m or 1.2 km.

c The distance is 6 full measures of the scale plus 1 cm more, giving 13 cm, so the real-life distance is 6 full measures (200 + 200 +200 + 200 + 200 + 200, which is 1200 m) plus the distance represented by 1 cm (which is 100 m) giving a total of 1300 m or 1.3 km.

7 Each 2 cm on the map represents 40 km on the ground.

a The distance on the map is 1 full measure of the scale or 2 cm, so the real-life distance is 40 km.

b The distance is 3 full measures of the scale or 6 cm plus 1 cm more, giving 7 cm., so the real-life distance is 3 full measures (40 + 40 +40, which is 120 km) plus the distance represented by 1 cm (which is 20 km) giving a total of 140 km.

c The distance is 3 full measures of the scale plus another $^1/_2$ cm more or 6.5 cm, so the real-life distance is 3 full measures (40 + 40 +40 which is 120 km) plus the distance represented by $^1/_2$ cm (which is 10 km) giving a total of 130 km.

8 1 cm on the map represents 100 m.

a The distance on the map is 6 cm, so the real distance 6 x 100 or 600 m.

b The distance on the map is 6 cm, so the real distance 6 x 100 or 600 m.

c The distance on map is 12 cm, so the real distance 12 x 100 or 1200 m or 1.2 km.

9 1 km on the ground is represented in 10 cm on the map so 1.3 km is represented in 13 cm on the map.

a The distance on the map is 4 cm to the closest part of the golf course and 11 cm to the furthest part of the golf course, so all the golf course is in mobile phone range.

b The distance on map is 15 cm, so the railway station is not in range.

c The distance on map is 7 cm, so the resort is in range.

d The distance on map is about 16 cm, so the sports field is not in range.

11 Measuring non-linear distance

1 There are more curves in the world than there are straight lines due to natural features.

2 1 cm on the map represents 100 m.

a The distance on map following the road is 7 cm, so the real life distance is 7 x 100 or 700m or 0.7 km.

b The shortest way is to travel along Beach Parade and turn left after the hospital. Proceed north to the road that is just past the Primary School. Turn right and continue in an easterly direction. Pass the kindergarten and the church. Just past the fire station is the resort. This is much shorter than following Beach Parade.

c The distance on the map is between 17 and 18 cm which is 1700 to 1800 metres or 1.7 to 1.8 km.

3 Yes

4 You could use a piece of string to follow the curves to get the distance on the map between 2 points. Mark the start and end points on the string. Then straighten the piece of string and measure the distance from your starting point to the end point. Use this distance and the scale to calculate the real-life distance.

5 Other things could be wire, wool, a small chain or anything that can follow the curves and then be straightened. A ruler could be used in the same way as paper. A pair of dividers could also be used to measure small stretches that are then plotted on an edge of paper, which is measured.

6 Some of the alleged lengths of the coastline of Australia are 19 658 km, 32 500 km, 36 735 km. 25 760 km and 132 000 km. The length obtained depends on the precision with which the measurement is made. The larger and more detailed the map, and the more finely the measurement is made, the longer will be the measured coastline.

7

a about 750 km

b about 1125 km

c about 875 km

d about 1250 km

e about 9000–10 000 km (also see table over)

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

A Typical set of results.

	1:20 000 000	1:80 000 000
Melb–Perth	2625	2500
Broome	1700	1500
Darwin	1125	1250
Brisbane	2725	2500
Melbourne	1375	1250
Total	9550	9000

f Measurements are more accurate the larger the scale of the map is.

8

a True **b** True **c** False **d** False
e True

12 Measuring area on maps

1 Tracing paper is used for the grid because it is see-through. It can be placed over the map and the feature can still be seen.

2 Some of the advantages are it doesn't harm the map and different size grids can be tried on different pieces of tracing paper. The grid can also be used on many different maps.

3

a 100 square kilometres
b 2500 square metres
c 25 square centimetres

4

a 100 square metres
b 500 square metres
c 2000 sq m

5

a 21 **b** 20 **c** 10 **d** 31 **e** 4
f 124 **g** 124 square kilometres

6

a 10 000 x 2 =20 000 square metres or 2 hectares
b 10 000 x 12 = 120 000 square metres or 12 hectares

7 One way of calculating the area of the reserve is to calculate the area covered by the reserve and the golf course together and then subtract the area of the golf course. Full area including reserve and golf course fits in 23 squares so reserve is 11 squares or 11 x 10 000 = 110 000 or 11 hectares. You could also just carefully follow the normal technique.

8 Your results should be almost the same. Some differences are likely as both ways give approximate results.

13 Representing heights on maps

1

a Many maps do not have any indication of height above sea level so the user gets a 'flat view'.
b It is useful to know about the heights of features and points on a map, particularly if there are steep hills or cliffs. Height is important to many users as it affects things like travel time between places. For example, a hiker using a map of a national park would need to know height.

2

a Point symbols locate the particular position of a feature. The spot height locates the position of a feature and as well as showing the height of the feature.
b Spot heights are usually used for prominent features such as mountains or deep parts of the sea.
c A spot height is a simple symbol and the height given is a single value that is correct for that point.

3 Hachures give a general guide to the shape of the land over a large area. Spot heights give an accurate height for a single point.

4

b There are seven elevation layers including one at sea level. There are three depth layers for the sea and a layer that shows land below sea level.
c Very low land that lies below the level of the sea. Sometimes it is a dry lake bed such as Lake Eyre or land reclaimed from the sea as in parts of Netherlands (Holland). It is sometimes described as a depression.
d Dark brown for land between 3000 and 4000 metres and purple for land over 4000 metres.
e Dark green for elevations up to 200 metres,
f The lightest blue is for the shallow water near the coast up to a depth of 200 metres.

5

a 100 metres
b Calculate by getting the difference between the heights of adjacent contour lines, such as 300 minus 200 equals 100.
c Contour interval is the vertical distance (height) between adjacent contour lines. The contour interval is the same for all contour lines on the same map.
d 4.5 cm
e 1.75 cm
f The right corner because contours are closer together over the distance.

6 **a** Between 760 and 900m.
b Between 2500 and 3800 m

7 **a** The feature is Mt. Whitehorn. It has a spot height of 2100 m.
c Local relief is 2100 (Mt. Whitehorn) minus 810 (airport) = 1290 m.

14 Finding things on maps

1 Houses appear on all +
a church **b** bowling alley **c** tent shop

2

a E1 or E2 or F2 are all correct. **b** B1 **c** F1
d C2 or D2 **e** D4 or E4

3

b It is called an alphanumeric grid because it uses letters of the alphabet as the labels in one direction and numbers as labels in the other.
f The black grid gives a general area of the features while the red grid only gives a broad area. A finer grid is easier to use to find a feature.

4 One answer is: Find where Blair Road crosses High Street. The church is to the west of Blair College.

5

a a square **c** a square
b a tick **d** a tick

15 Area references to grid references

1 The labels on the bottom grid lines should be the same as the labels on the top grid lines. The labels on the right grid lines should be the same as the labels on the left grid lines. A grid line only makes sense if the label on one end of the line is the same as the label on the other end of the line.

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

2 The system is a grid coordinate system. Four-figure area references could be used to locate things on the map. The area reference defines the location within one grid square.

3

a The area references are:
Lookout 8433
Houses 8231
Lake 8131

b The houses and lake cover more than one grid square. Where the feature covers more than one grid square, the area reference used should be for the middle of the feature.

c See answer 3a above.

4

a 30 mm by 30 mm

b 3 mm

5

b 331 and 339

c 845 and 335

6 Middle house is at 824318. Middle of the lake is at 813319.

7 900 m approximately

9 Slight variations are possible

a 606302

b 542282

c 586256

d 555310

16 Mapping a tribal group area

Ask students to swap their answers with a classmate for checking or have a class discussion on the answers.

17 Mapping ancient Athens

1 The main map is of Athens and Piraeus. The inset map is a small map of a much larger area that includes Athens. The inset map shows where Athens fits in a larger geographical area. The inset map is a small-scale map as it shows a relatively large area on the ground.

2 Piraeus is 9–10 km south-west of Athens. The Long Walls provide a safe protected route for people and goods moving between the two places. Probably the two towns were friends but they needed protection from nearby enemies. As Athens was inland and Piraeus was on the coast, one of the items of trade may have been fish and other seafood. If Athens' traders wanted to sell their goods to other places they may have used ships leaving from Piraeus to transport their goods.

3

a From east to west is about 1700 metres or 1.7 km. From north to south is about 1500 metres.

b The approximate area of the walled city is 2.1 square kilometres.
Techniques could be:

- Create a grid over the map (or using tracing paper) with sides covering 250 metres of real distance (using the scale for the size of the grid squares). Each square would cover 1/16 or .0625 of a square kilometre. Count the number of grid squares that fall completely within the walled city. Count the number of grid squares that are partly within the walled city, Take half of this number and add to number of the full squares. Multiply the area of one square by this number to give the total area.
- Use the same approach but make the grid squares equivalent to 100 metres of real distance.
- Cut out around the walled cities. Make a rectangular shape by cutting off pieces of the map that jut out and use them to fill in the holes in the shape until the shape is a rectangle, Measure the length and width of the final rectangle and use the scale to calculate the area.

c The approximate length of the wall is 7.2 km.
Techniques could be:

- Run a piece of string around the wall. Get the length of the string used and use the scale to calculate the real length,
- Instead of string, find the length of the wall on the map using a piece of paper and mark at each change of direction or use a ruler or a piece of wire. The general approach is to get as accurate length of the wall on the map as you can and then use the scale to convert to real length.

4

a Piraeus

b The road runs between walls so there is protection from enemies.

d The roads start in a gate of the walled city. The same thickness of line is used for all the roads. The roads also have an arrow showing the direction to the named destination. There are 8 other roads leaving the walled city and another road branching off the road to Academy that leads to Piraeus.

5

a Hachures are used to show the hills.

c Local relief of the Acropolis in relation to Athens is 80 metres.

d The wall would be an extra line of defence, particularly if the Piraeusians became unfriendly. It may provide a good lookout to the south from the Hill of the Muses. It also gives a barrier to easy access between the Hill of the Nymphs and the Hill of the Muses.

18 Different views of a city

1 Dark grey

2 Central Bus depot, Central Station

3 As it is not in the key, you need to recognise it by the shapes of the sporting grounds.

4 Parks and gardens and government.

5 To the south, the land-use zones are the parks and gardens, arts and culture, and residential apartments. To the north the land uses are CBD (business, commercial, shopping) and transport.

8 Some of the things are the sports grounds in the sporting precinct, the parks and gardens precinct, some of the smaller buildings north of the tall buildings and a better view of the skyline of the city.

9 Some are the facades of the buildings, the train at Central Station, the bus leaving the bus depot and the smaller buildings in the streetscape.

19 Multiple choice test

1 d	**10** d	**19** b
2 c	**11** a	**20** d
3 c	**12** d	**21** c
4 d	**13** a	**22** d
5 b	**14** b	**23** b
6 a	**15** c	**24** a
7 d	**16** d	**25** c
8 c	**17** b	**26** b
9 b	**18** a	**27** d

© Blake Education ISBN 978 1 92561 035 2 Middle Years Geography Bk 1

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities that can be photocopied for student use. They provide a range of high-quality learning materials.

This checklist highlights the major uses and benefits of these books.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Middle Years Geography Book 1

This book is part of a two-book series that provides teachers with innovative lessons on topics which teach students essential geography skills. It will suit students at the middle years of schooling (years 5-9) and provide information and activities not found in core textbooks. Book 1 explores the basic elements of all maps. Sample answers have been provided where appropriate to make marking of student work easy and time-efficient.

Concepts covered include:

- Different ways of viewing the world
- The use of symbols and keys
- Compass direction
- Compass bearings
- Map scale and maps at different scales
- Measuring distances along straight lines and around curves
- Travelling time
- Techniques for measuring the area of features
- Techniques for representing height and depth
- Techniques for locating features
- Grid systems
- Mapping three different communities throughout time

Associated titles in the Instant Lessons series

- Middle Years Geography Book 2: understanding atlases and map types
- Geography Essentials: understanding diagrams
- Geography Essentials: understanding data and graphs

Blake EDUCATION
Better ways to learn

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

ISBN 978-1-92561-035-2